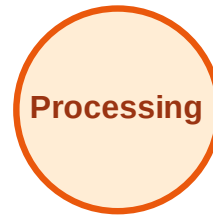


BFS Traversal

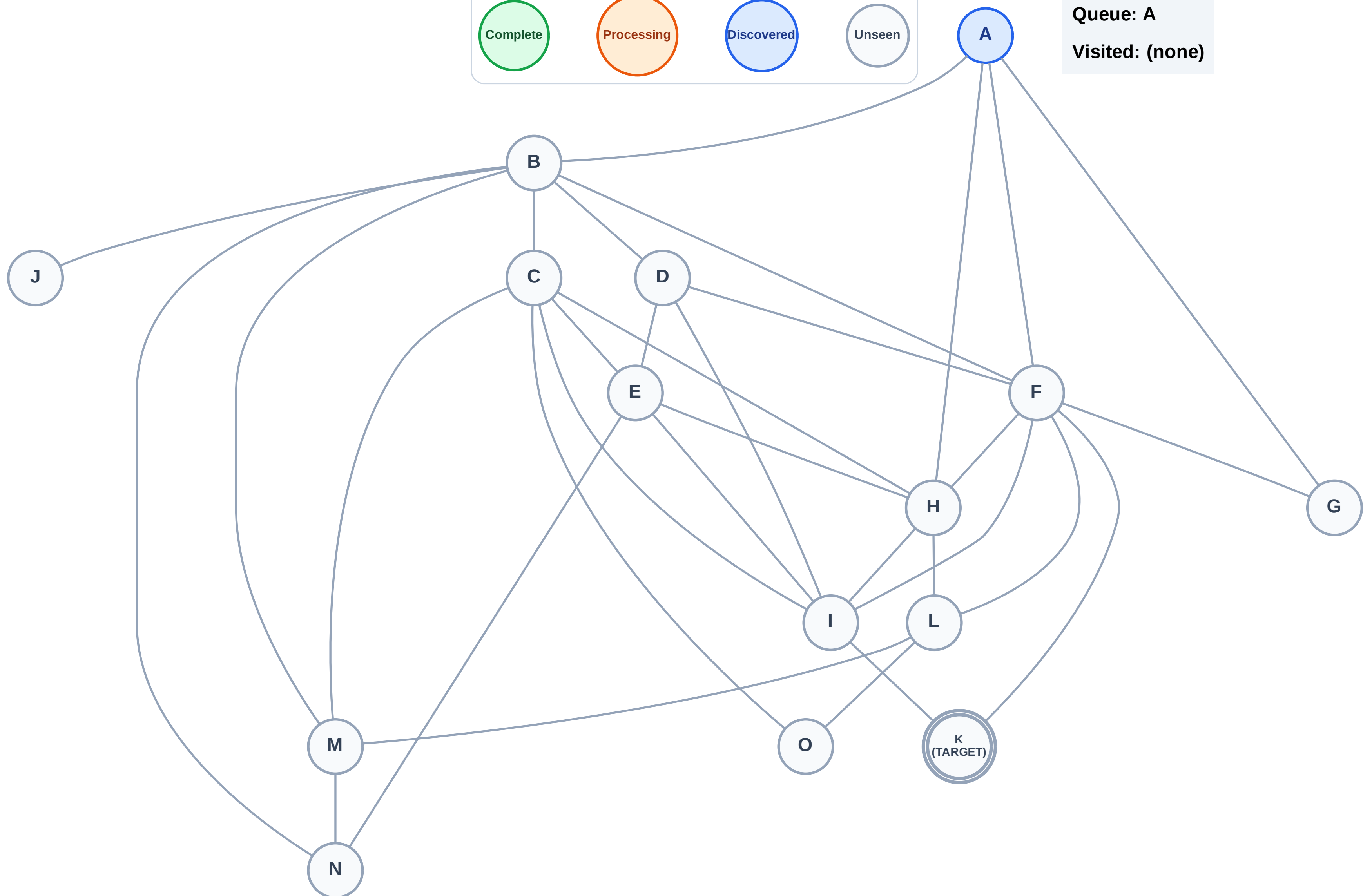
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



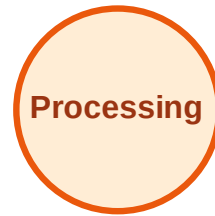
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



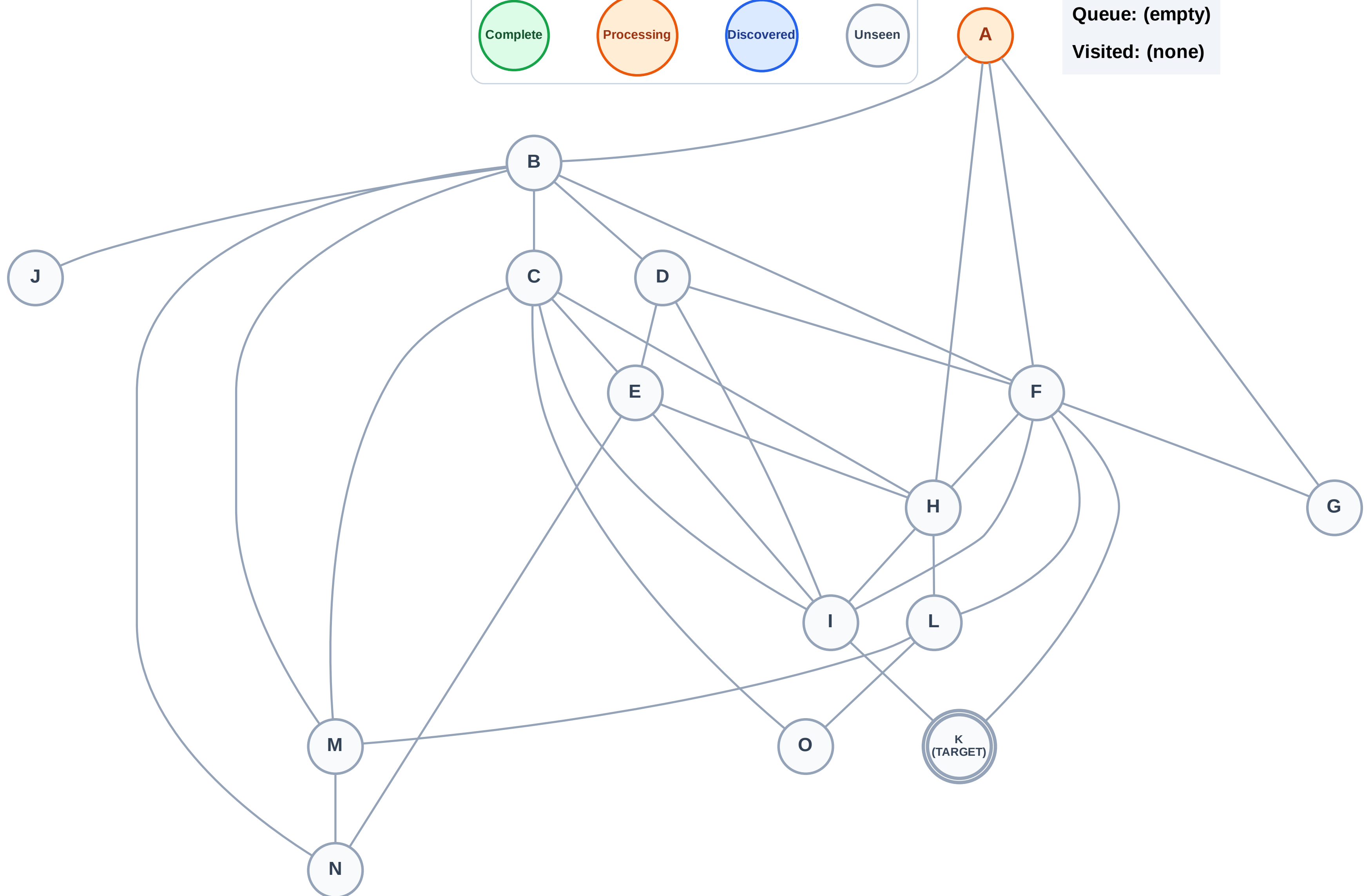
Discovered



Unseen

Queue: (empty)

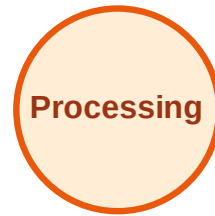
Visited: (none)



BFS Traversal

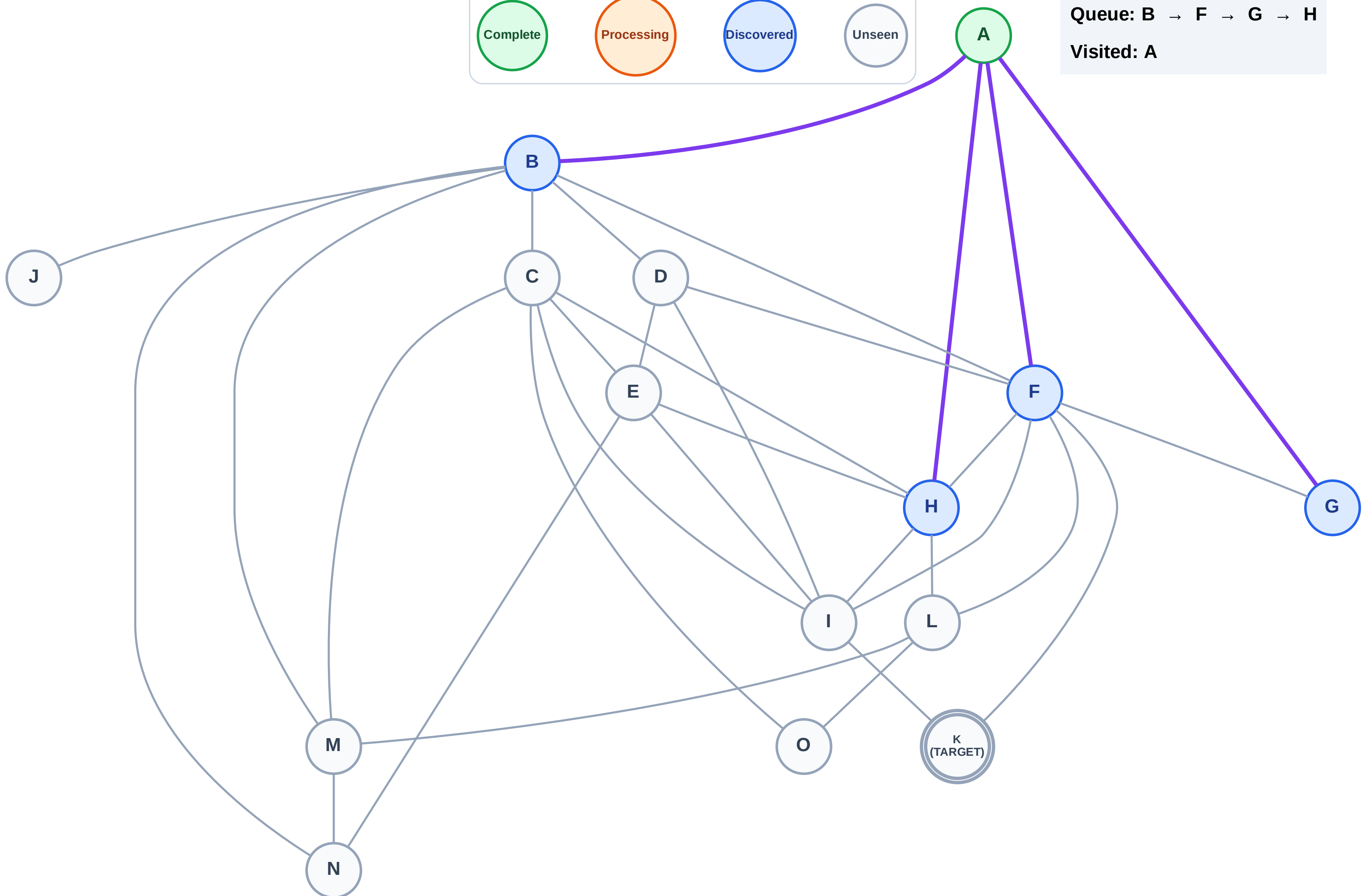
Step 3: Discover B, F, G, H from A

Legend



Queue: B → F → G → H

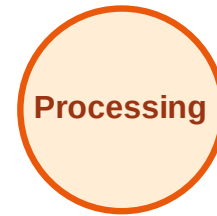
Visited: A



BFS Traversal

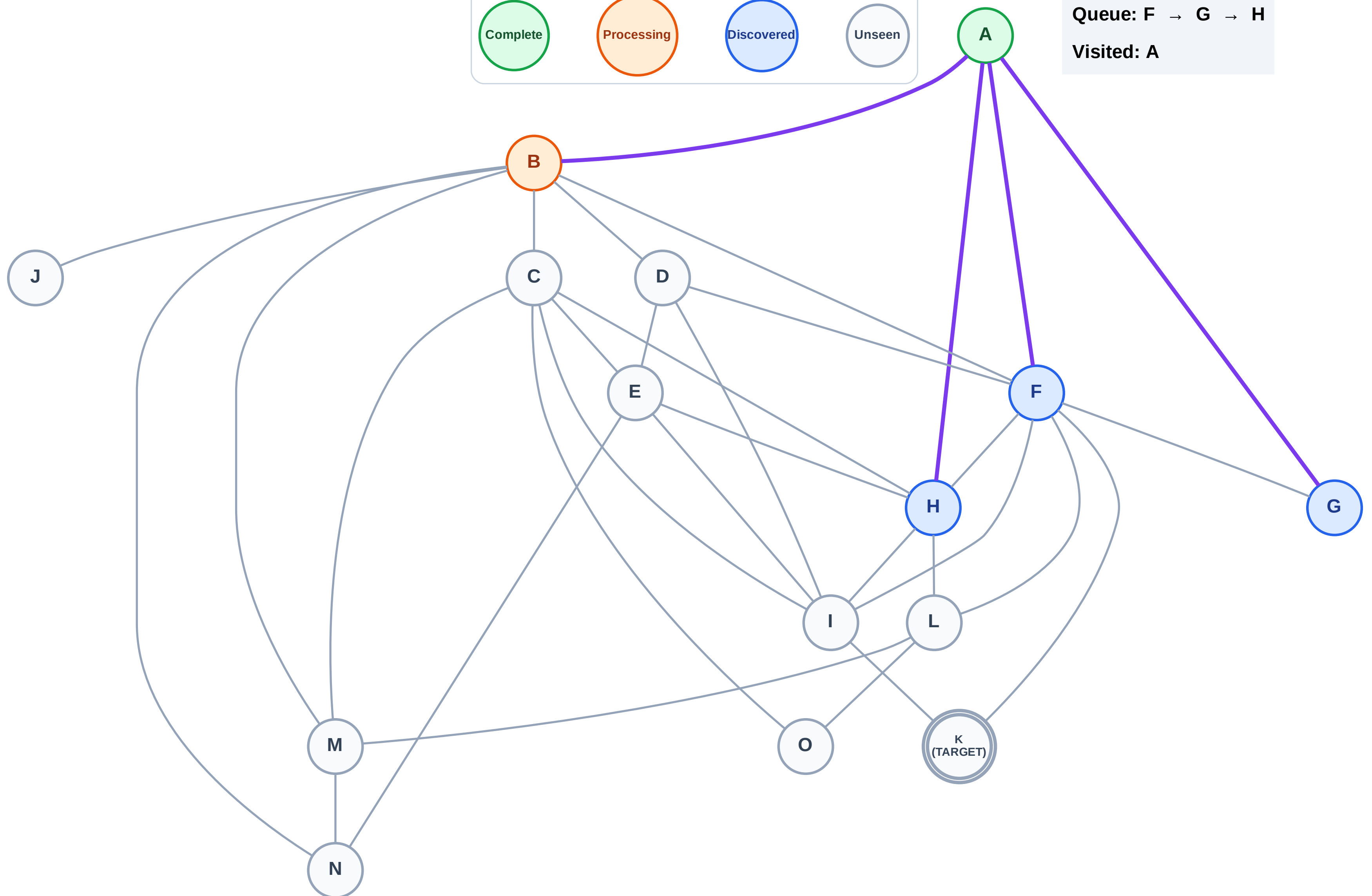
Step 4: Process node B

Legend



Queue: F → G → H

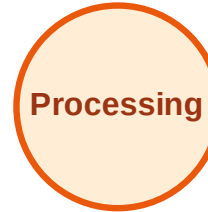
Visited: A



BFS Traversal

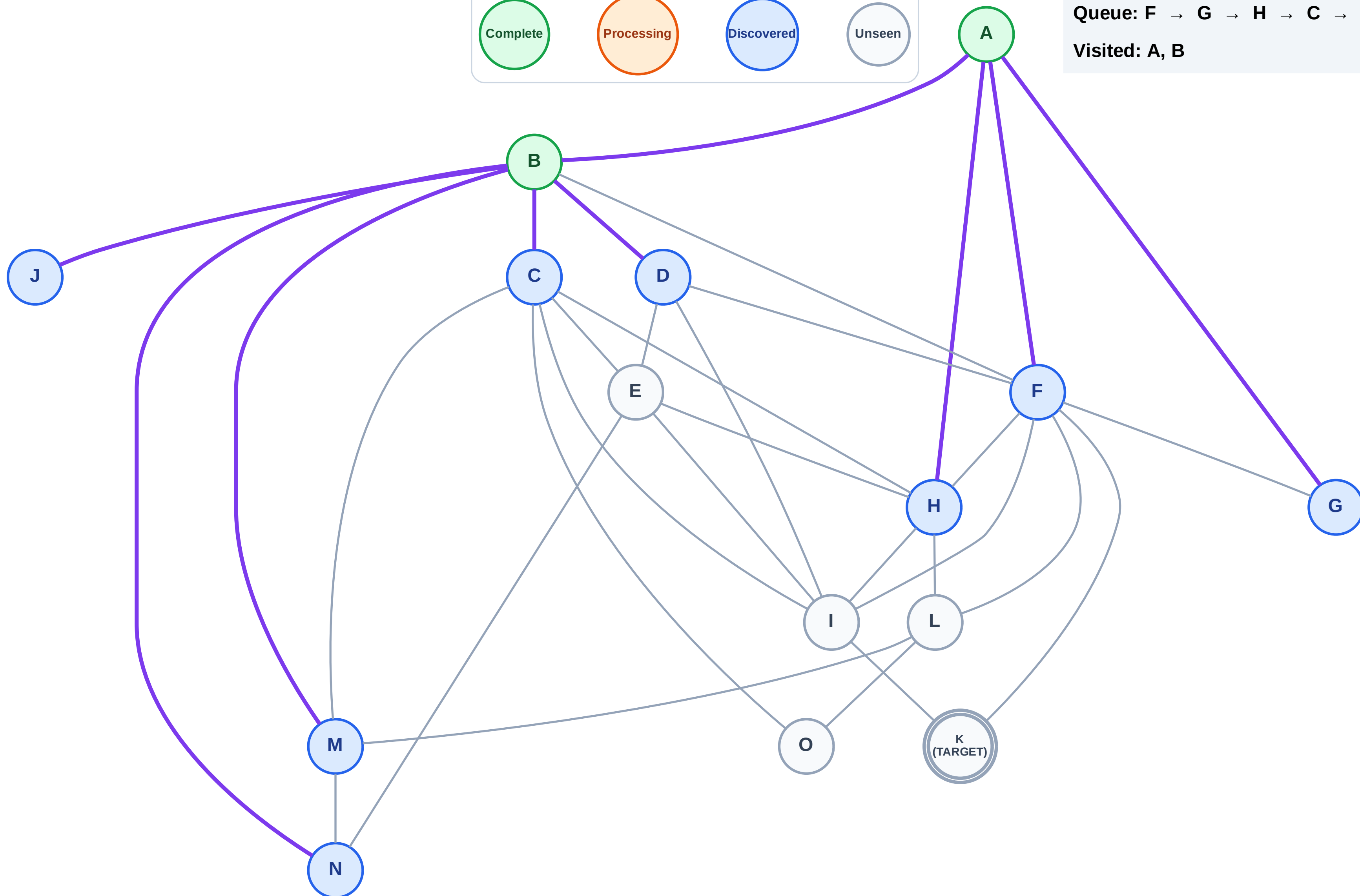
Step 5: Discover C, D, J, M, N from B

Legend



Queue: F → G → H → C → D → J → M → N

Visited: A, B



BFS Traversal

Step 6: Process node F

Legend

Complete

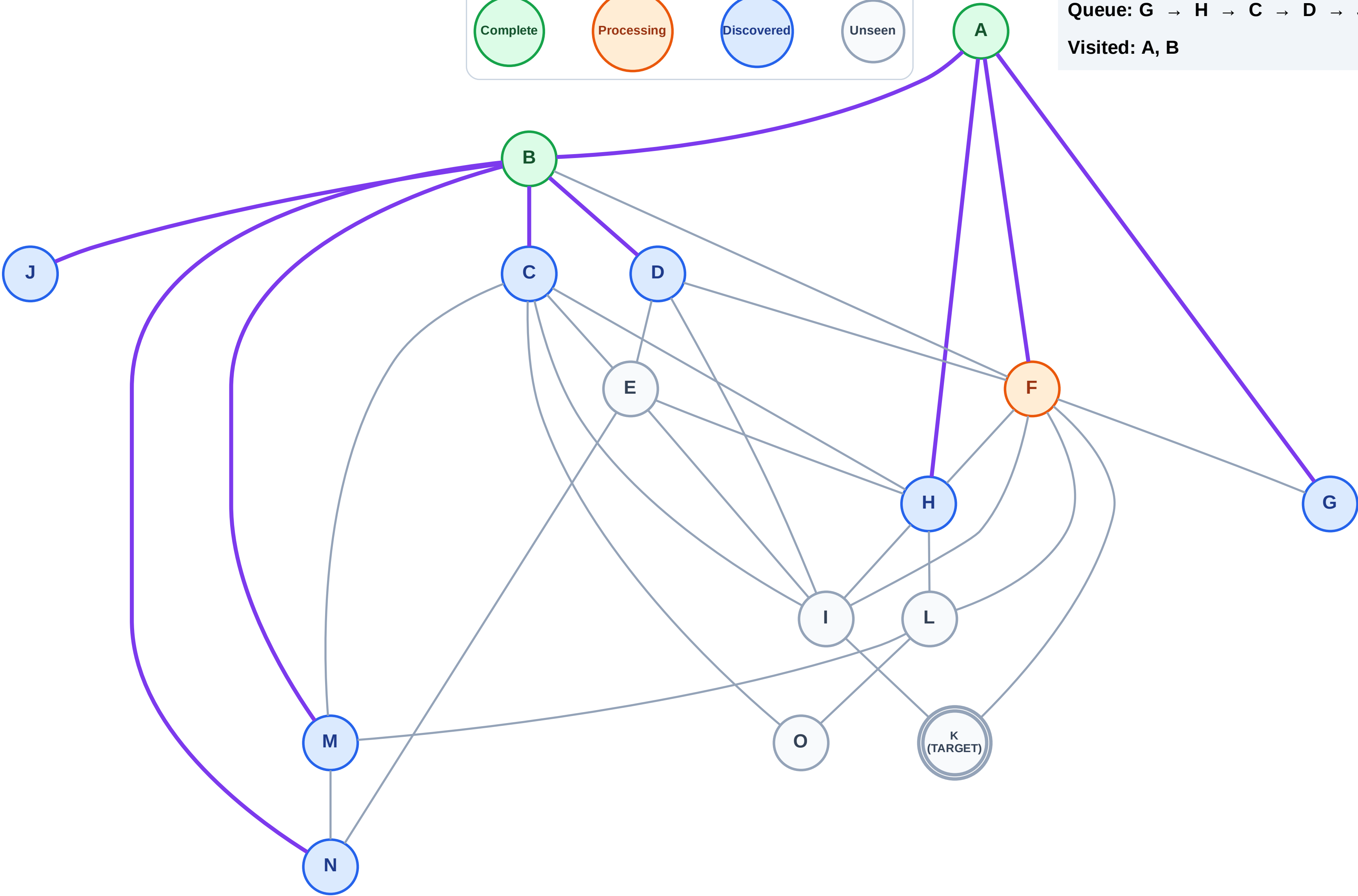
Processing

Discovered

Unseen

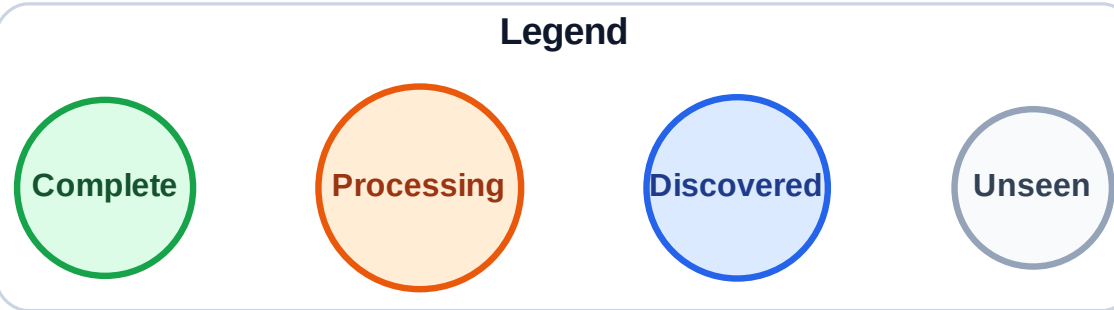
Queue: G → H → C → D → J → M → N

Visited: A, B

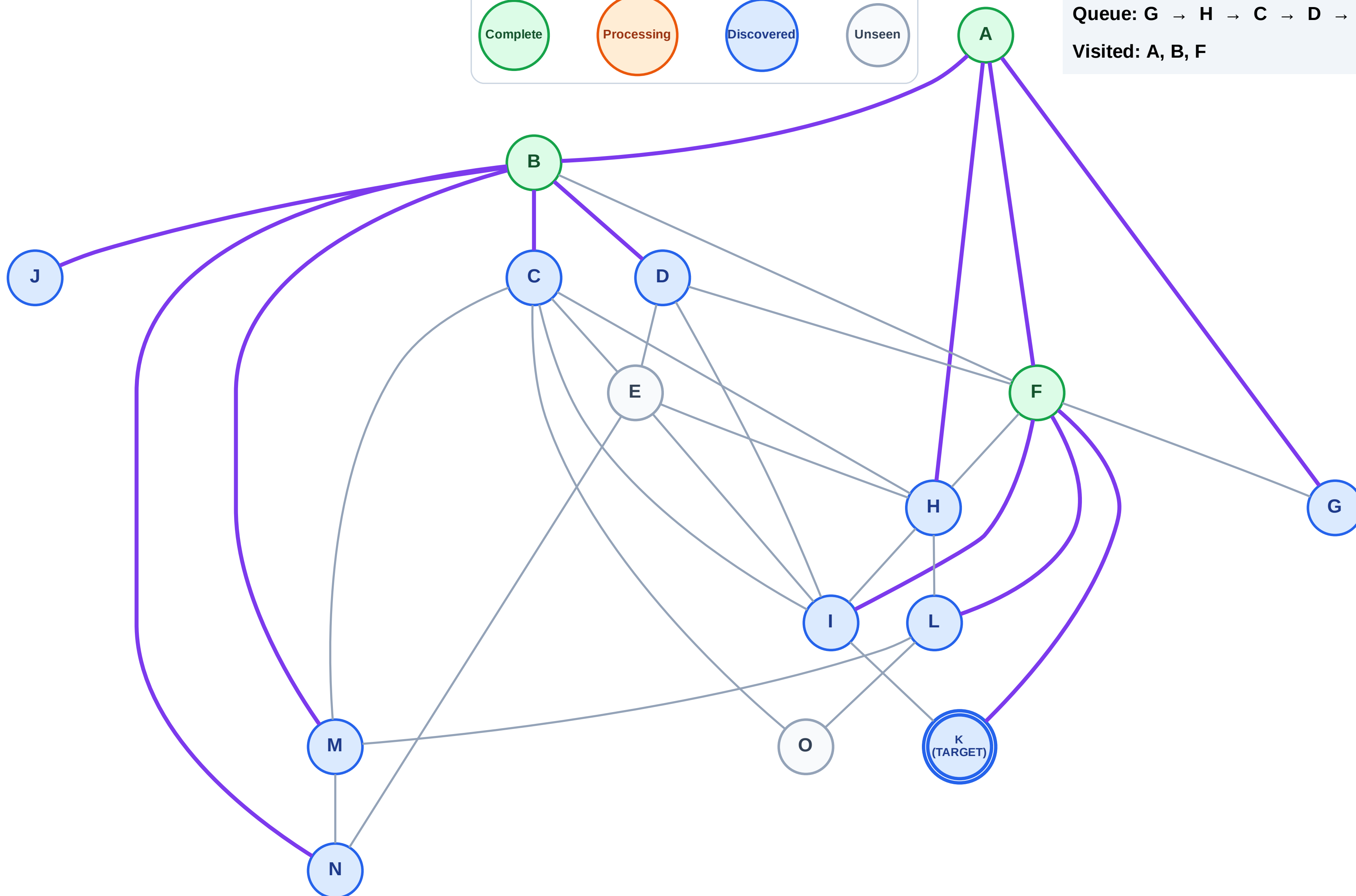


BFS Traversal

Step 7: Discover I, K, L from F



Queue: G → H → C → D → J → M → N → I → K → L
Visited: A, B, F



BFS Traversal

Step 8: Process node G

Legend

Complete

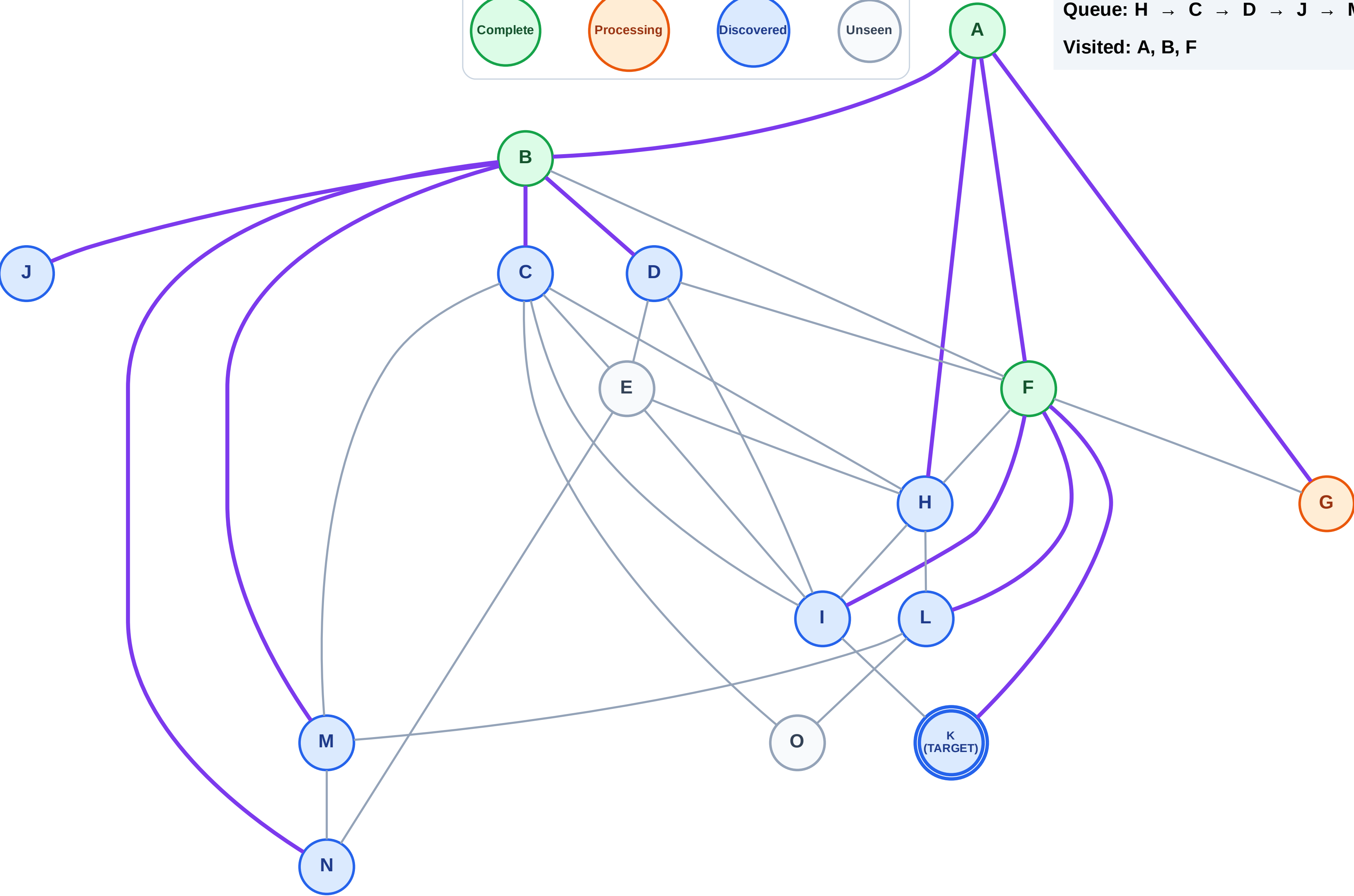
Processing

Discovered

Unseen

Queue: H → C → D → J → M → N → I → K → L

Visited: A, B, F



BFS Traversal

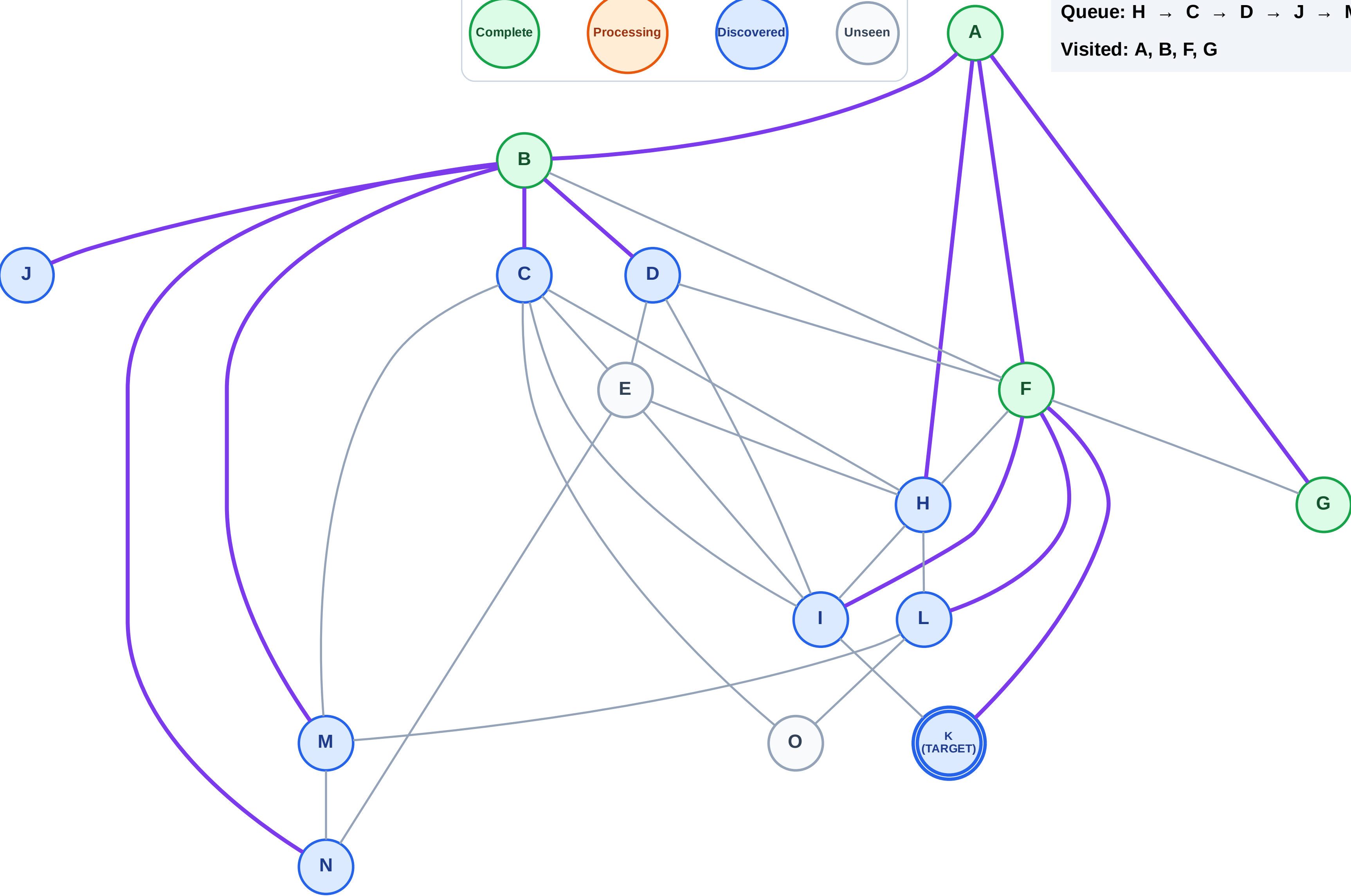
Step 9: No new neighbors from G

Legend



Queue: H → C → D → J → M → N → I → K → L

Visited: A, B, F, G



BFS Traversal

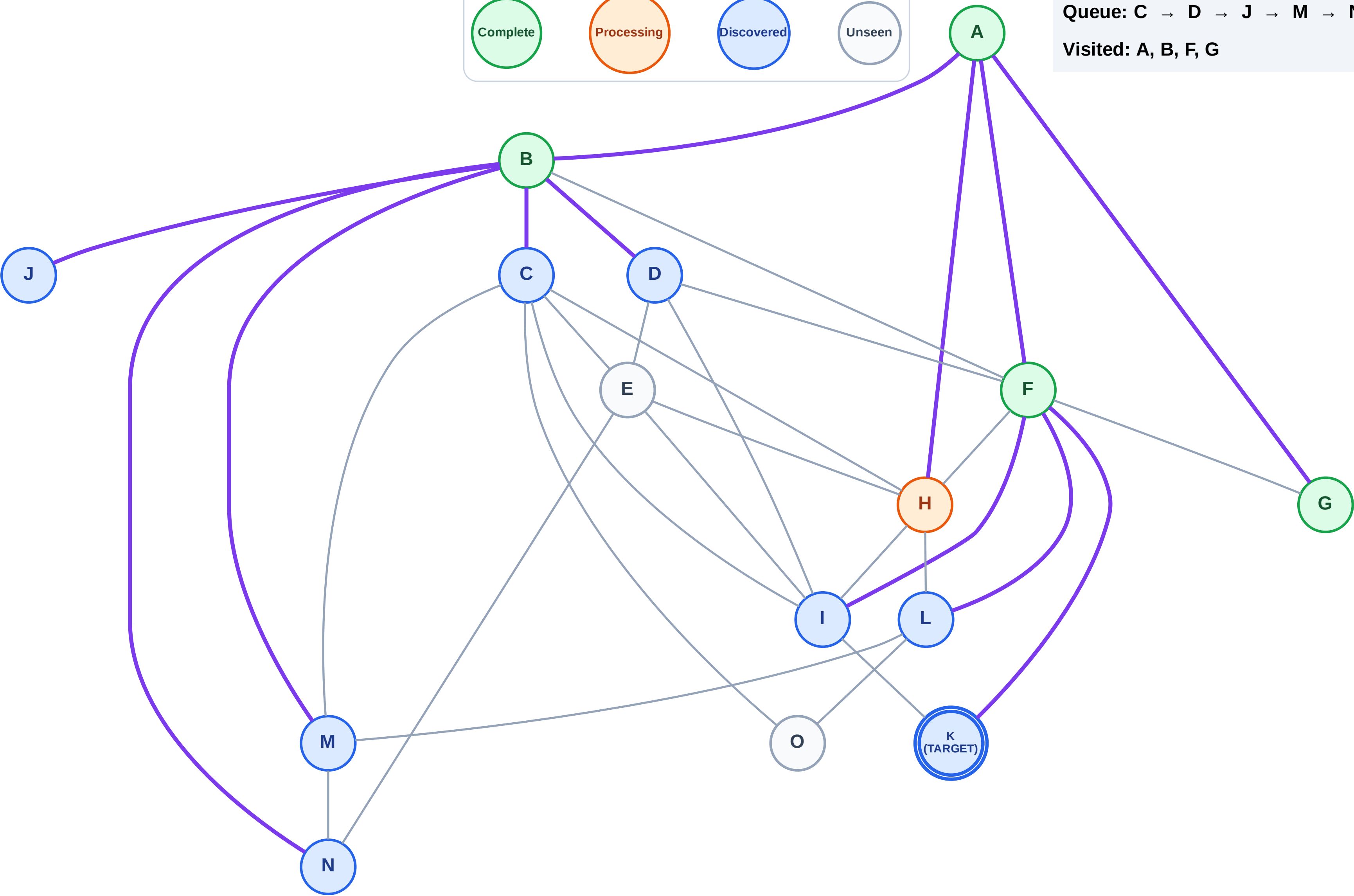
Step 10: Process node H

Legend



Queue: C → D → J → M → N → I → K → L

Visited: A, B, F, G



BFS Traversal

Step 11: Discover E from H

Legend

Complete

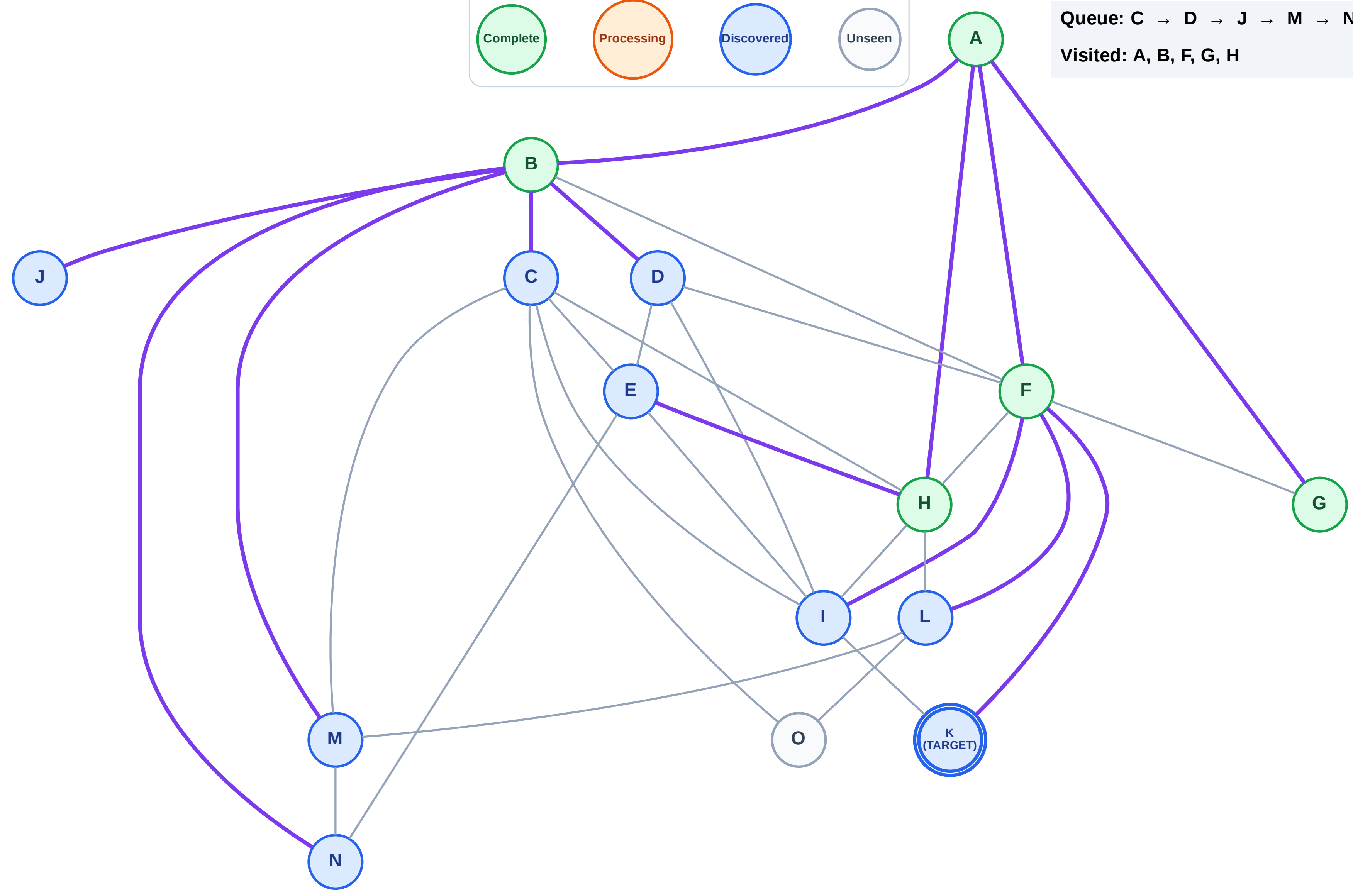
Processing

Discovered

Unseen

Queue: C → D → J → M → N → I → K → L → E

Visited: A, B, F, G, H



BFS Traversal

Step 12: Process node C

Legend

Complete

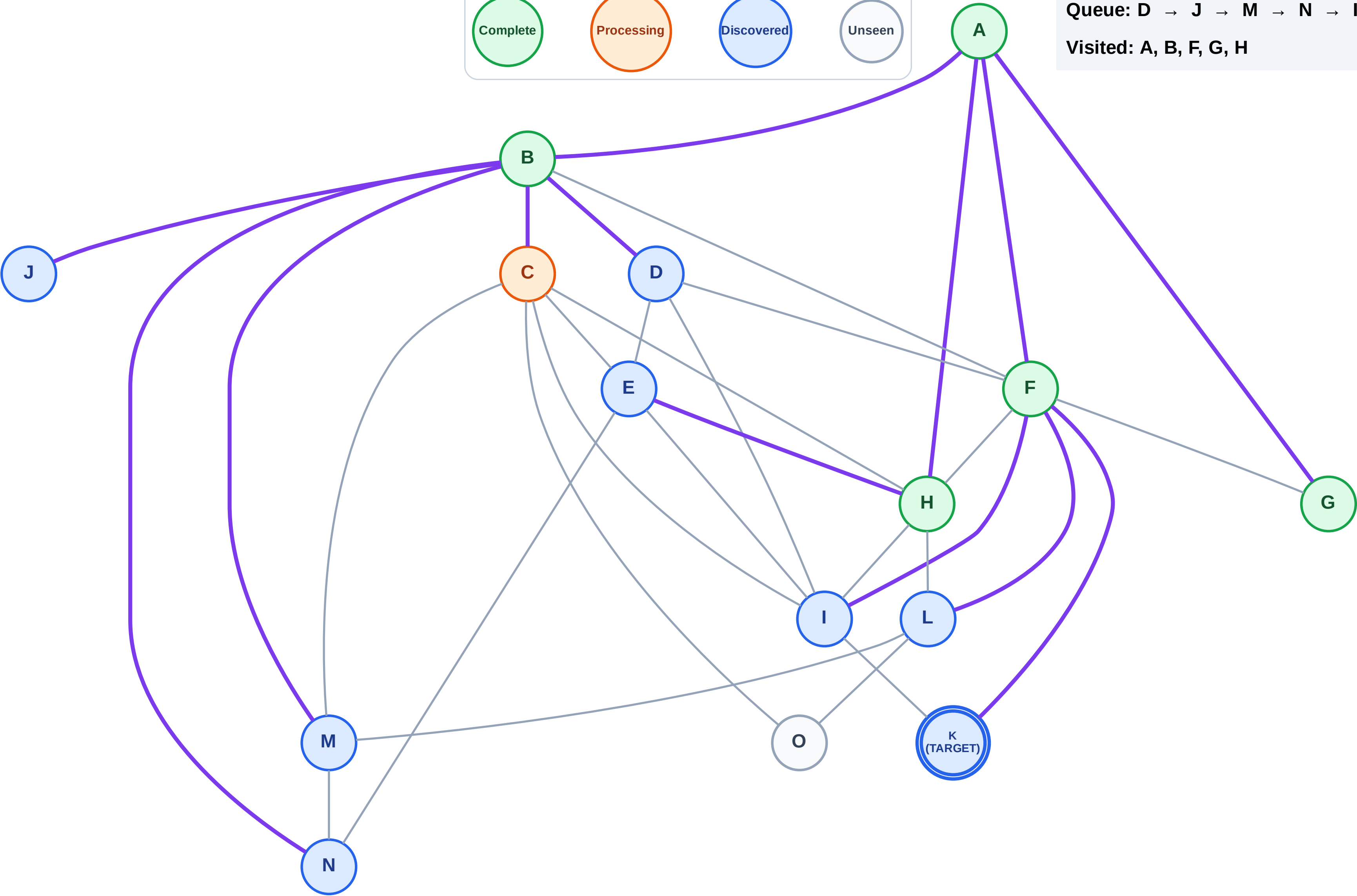
Processing

Discovered

Unseen

Queue: D → J → M → N → I → K → L → E

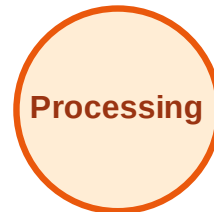
Visited: A, B, F, G, H



BFS Traversal

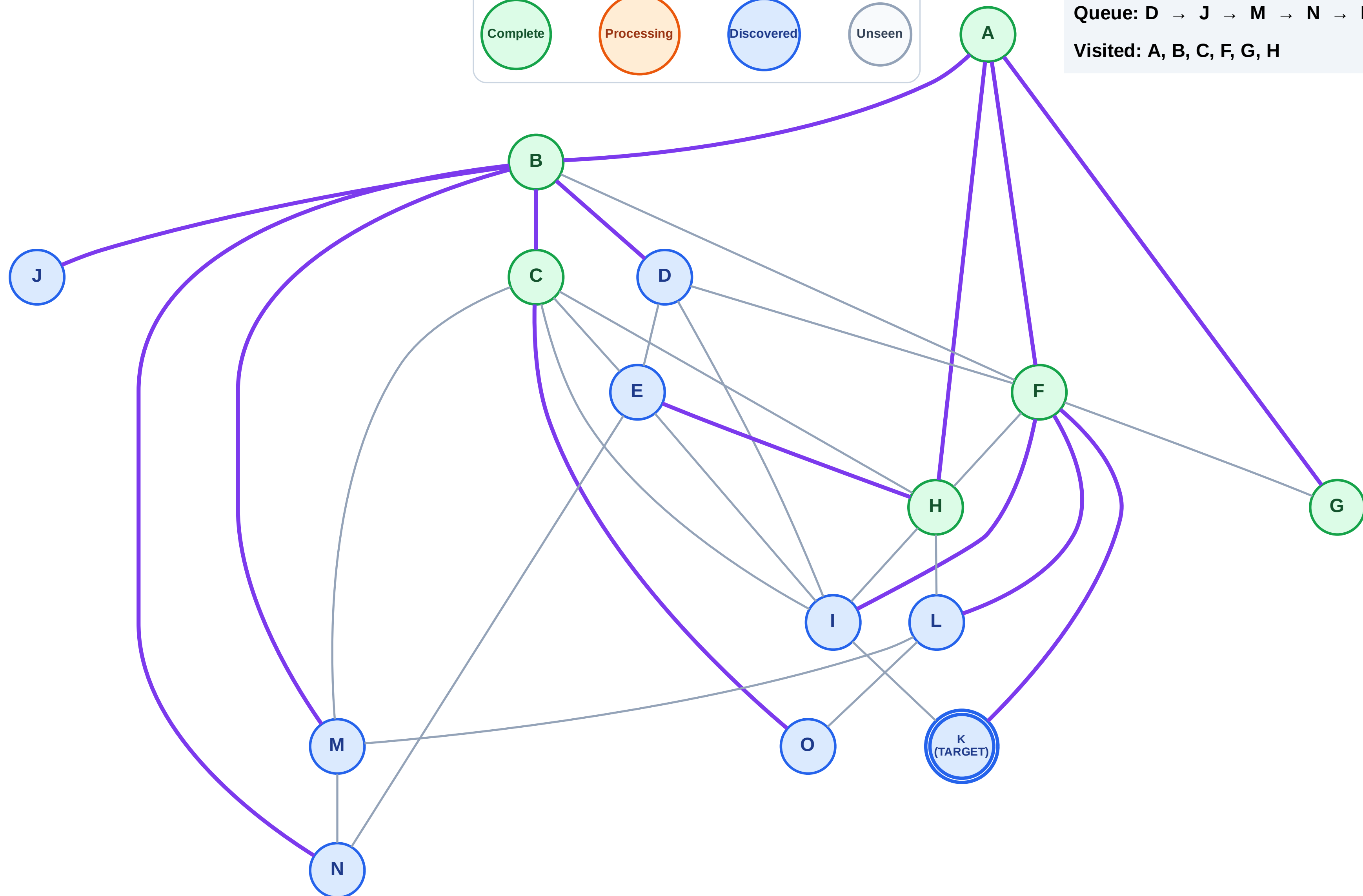
Step 13: Discover O from C

Legend



Queue: D → J → M → N → I → K → L → E → O

Visited: A, B, C, F, G, H



BFS Traversal

Step 14: Process node D

Legend

Complete

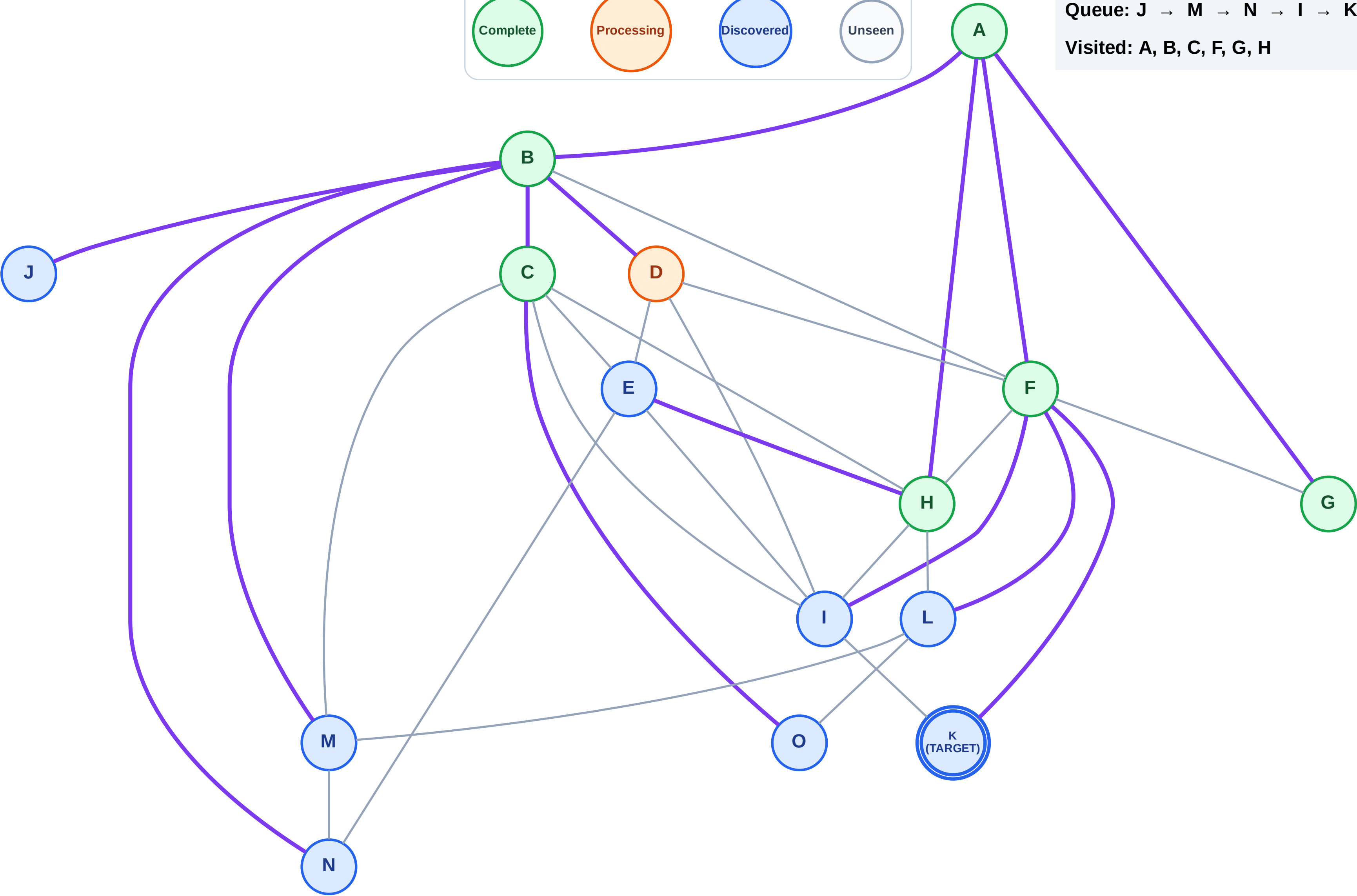
Processing

Discovered

Unseen

Queue: J → M → N → I → K → L → E → O

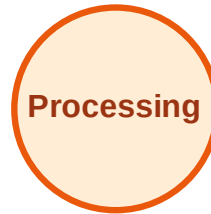
Visited: A, B, C, F, G, H



BFS Traversal

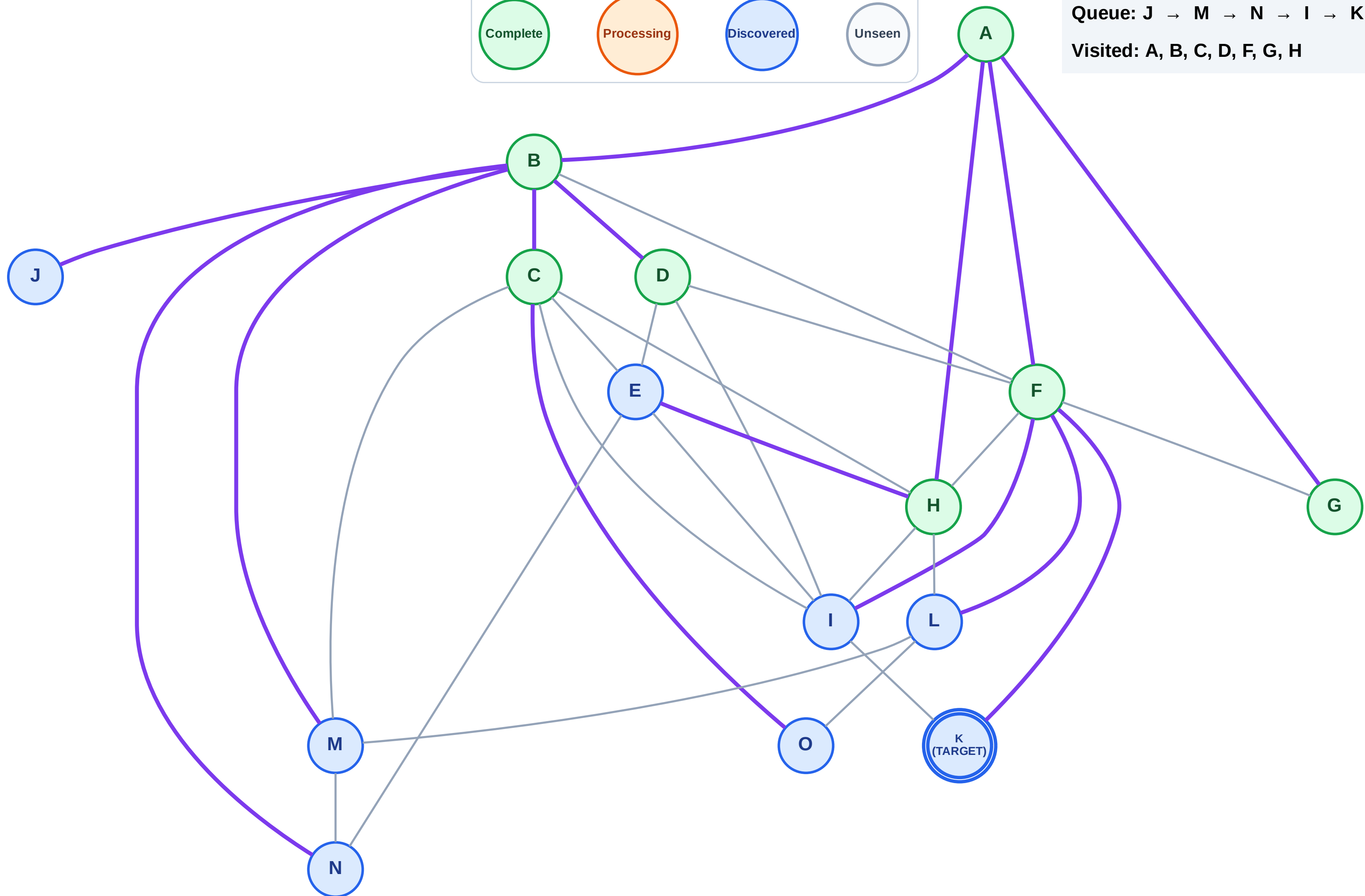
Step 15: No new neighbors from D

Legend



Queue: J → M → N → I → K → L → E → O

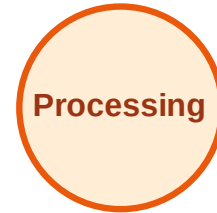
Visited: A, B, C, D, F, G, H



BFS Traversal

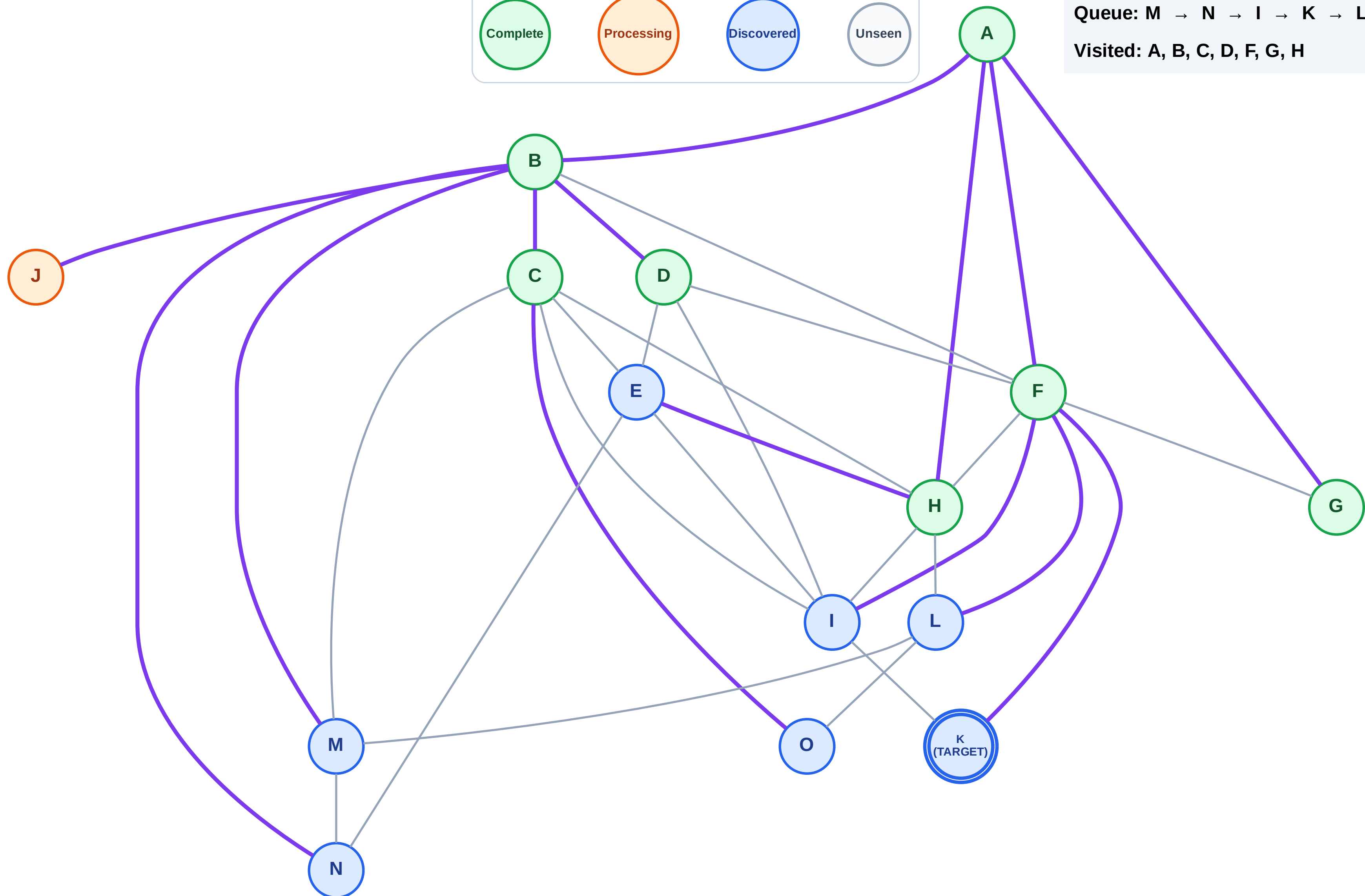
Step 16: Process node J

Legend



Queue: M → N → I → K → L → E → O

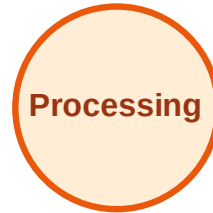
Visited: A, B, C, D, F, G, H



BFS Traversal

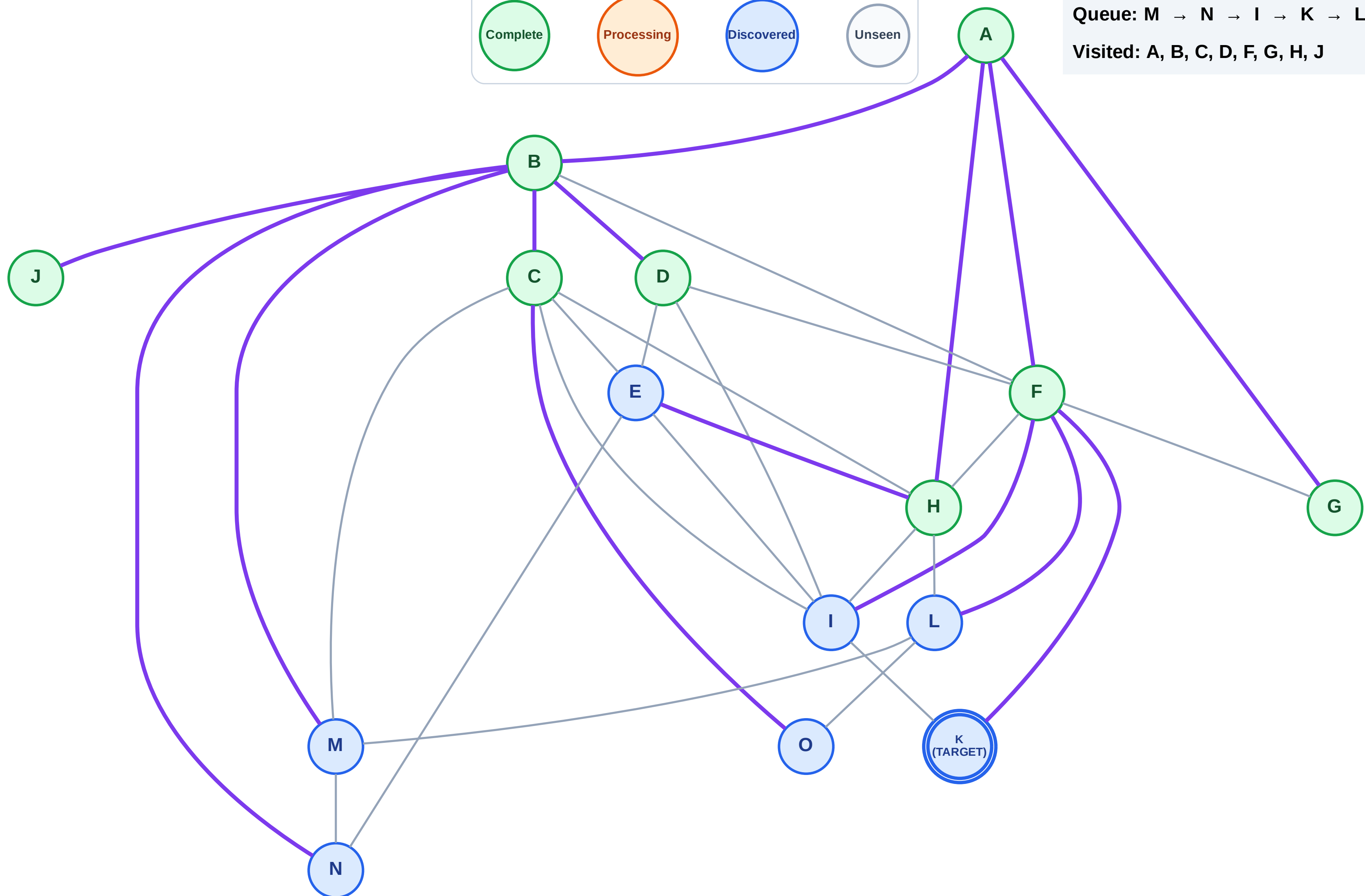
Step 17: No new neighbors from J

Legend



Queue: M → N → I → K → L → E → O

Visited: A, B, C, D, F, G, H, J



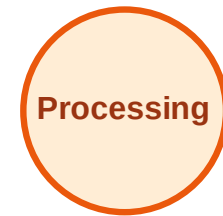
BFS Traversal

Step 18: Process node M

Legend



Complete



Processing



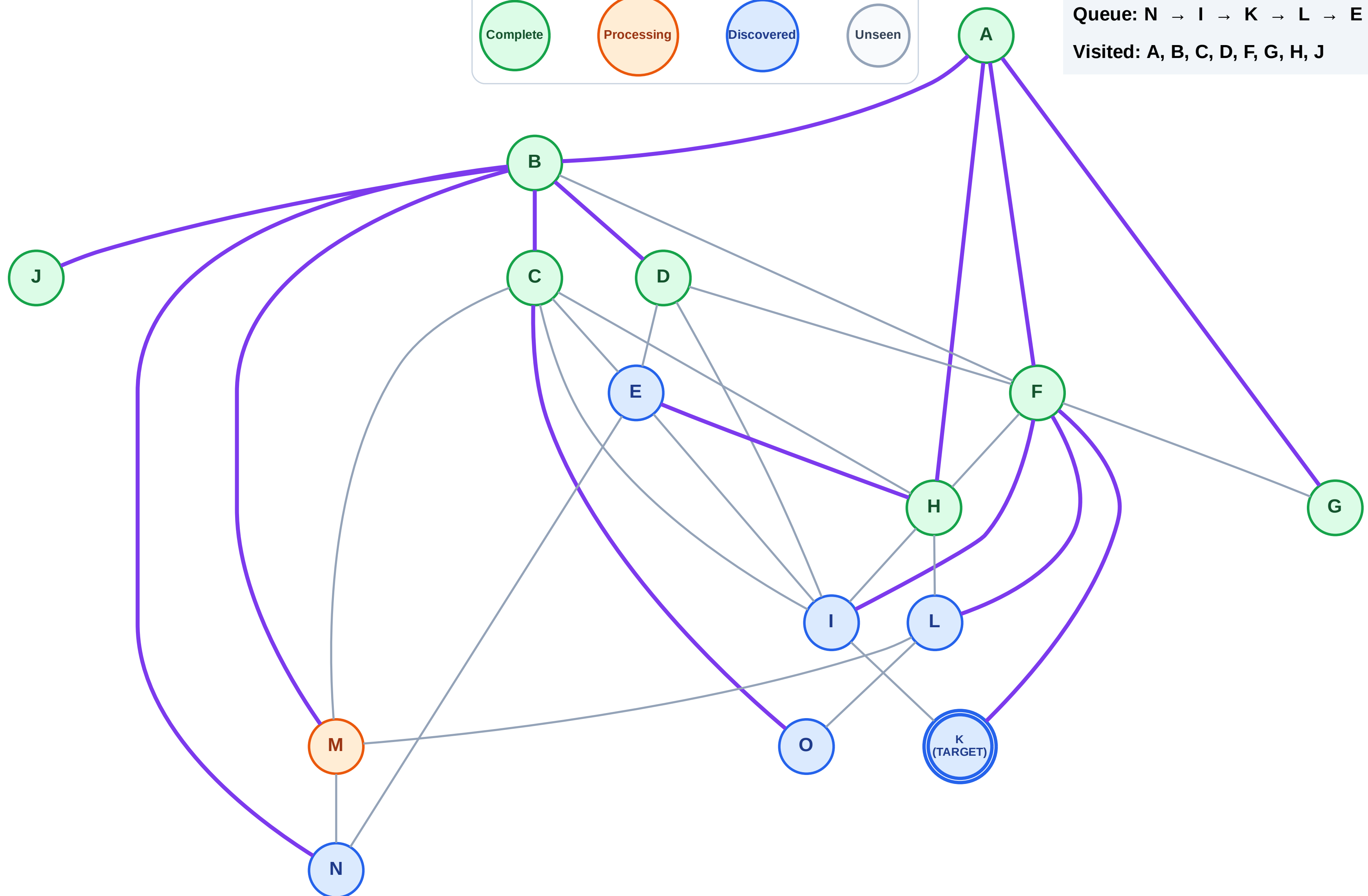
Discovered



Unseen

Queue: N → I → K → L → E → O

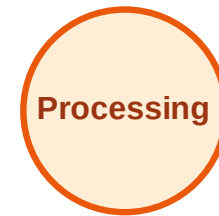
Visited: A, B, C, D, F, G, H, J



BFS Traversal

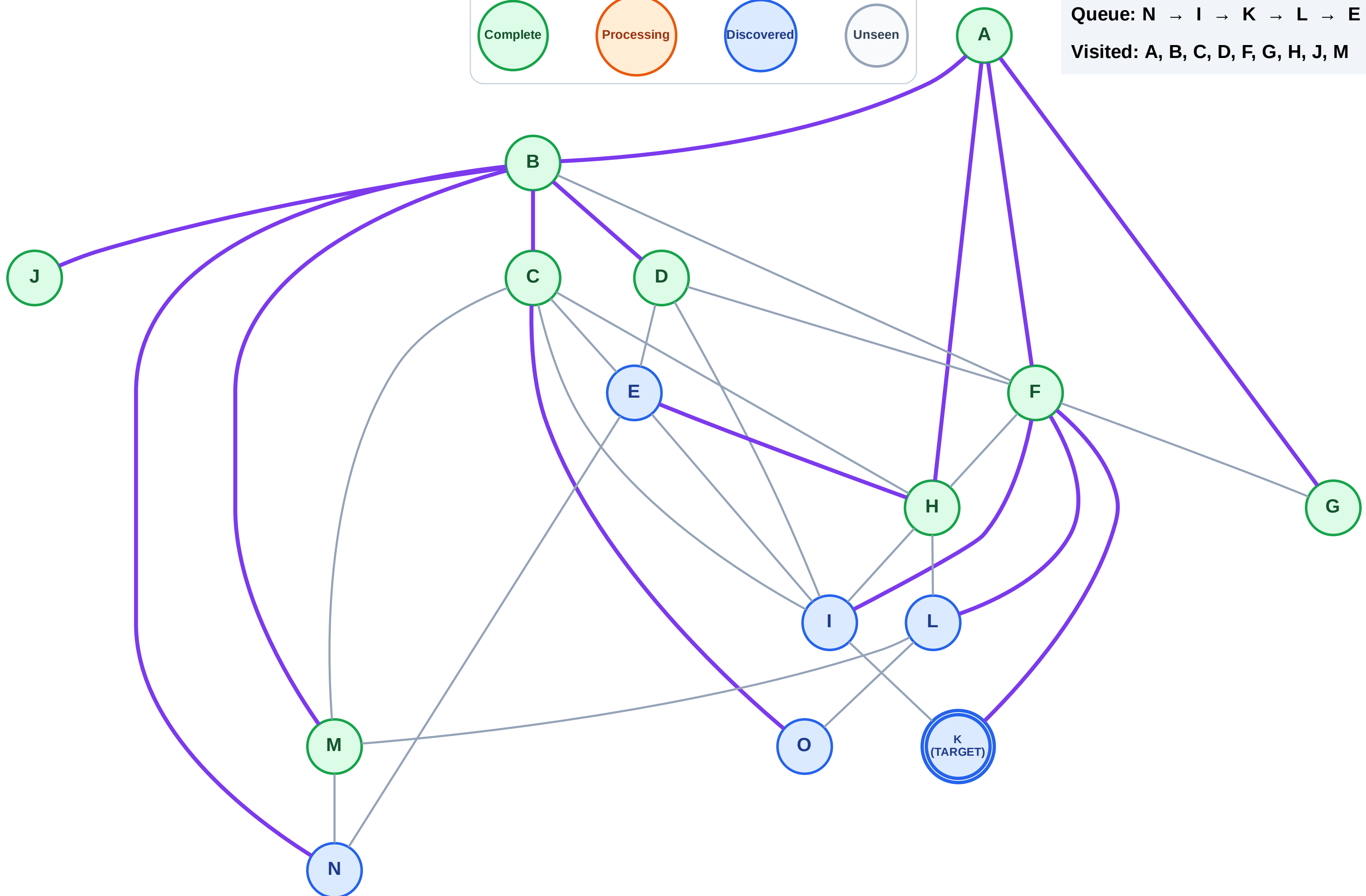
Step 19: No new neighbors from M

Legend



Queue: N → I → K → L → E → O

Visited: A, B, C, D, F, G, H, J, M



BFS Traversal

Step 20: Process node N

Legend

Complete

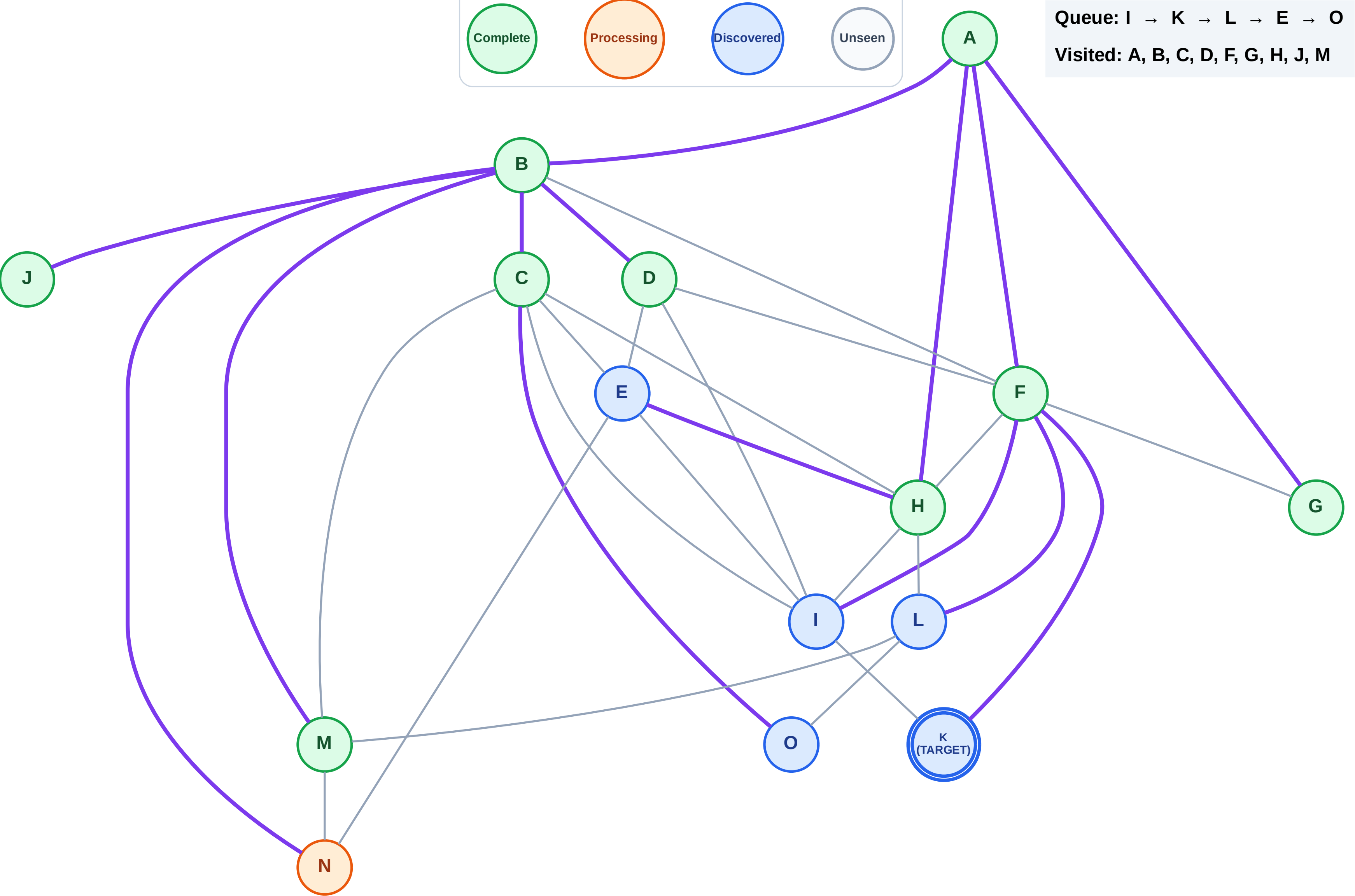
Processing

Discovered

Unseen

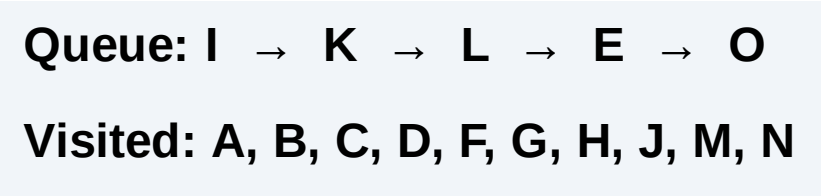
Queue: I → K → L → E → O

Visited: A, B, C, D, F, G, H, J, M



Step 21: No new neighbors from N

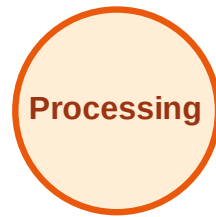
Step 21: No new neighbors from N



BFS Traversal

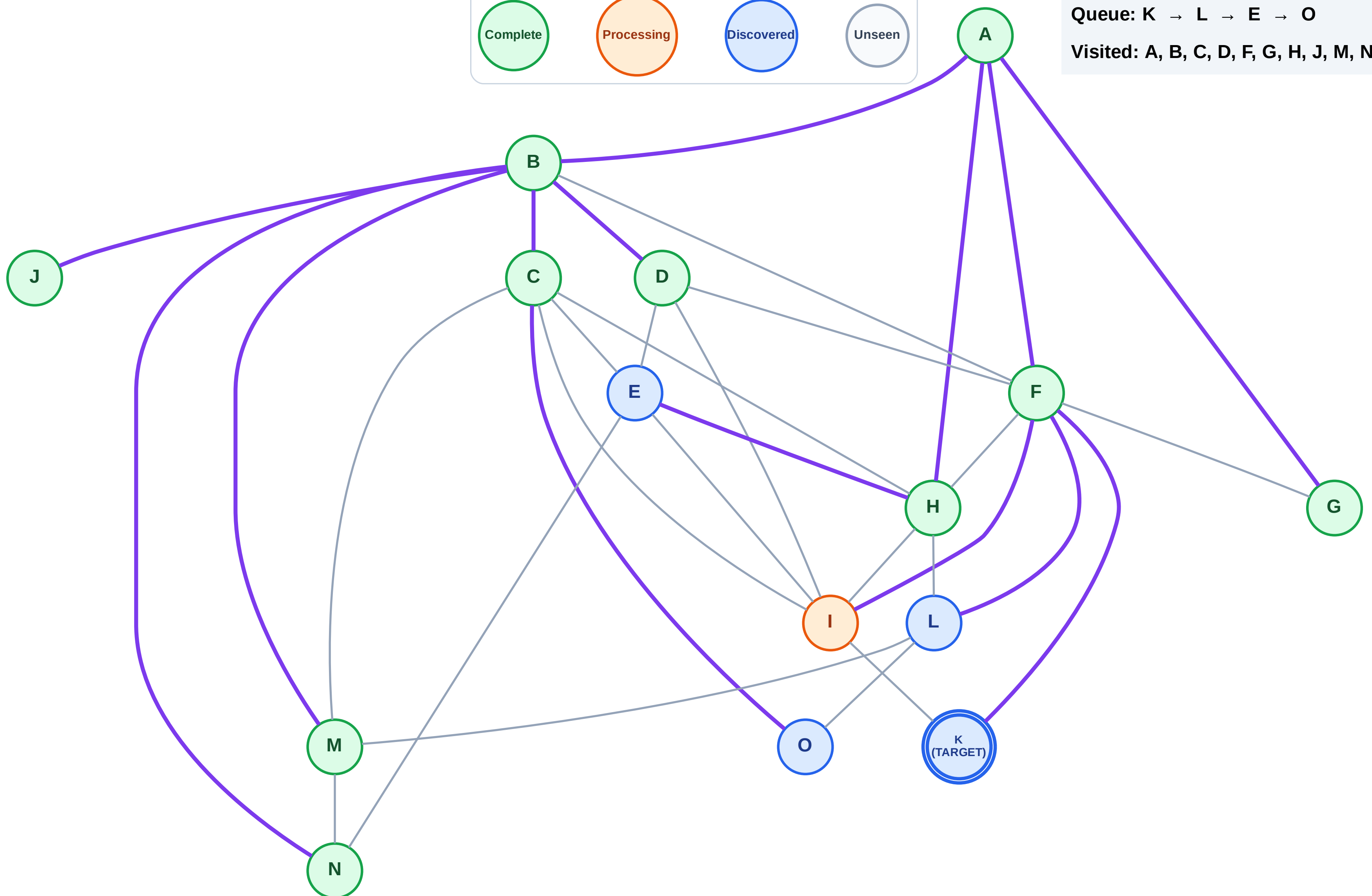
Step 22: Process node I

Legend



Queue: K → L → E → O

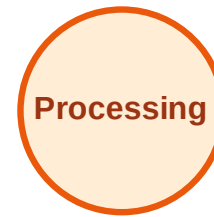
Visited: A, B, C, D, F, G, H, J, M, N



BFS Traversal

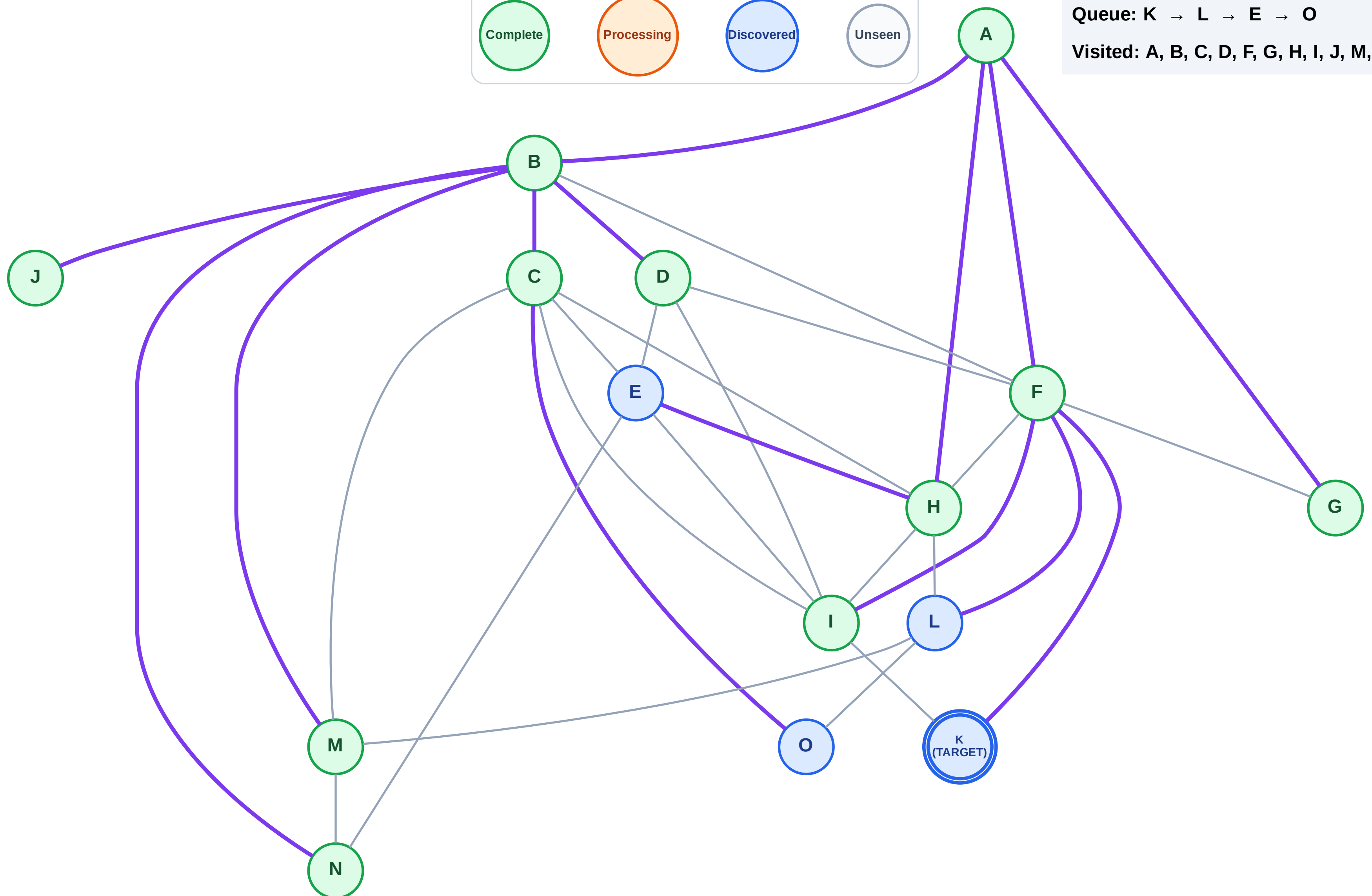
Step 23: No new neighbors from I

Legend



Queue: K → L → E → O

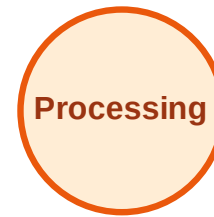
Visited: A, B, C, D, F, G, H, I, J, M, N



BFS Traversal

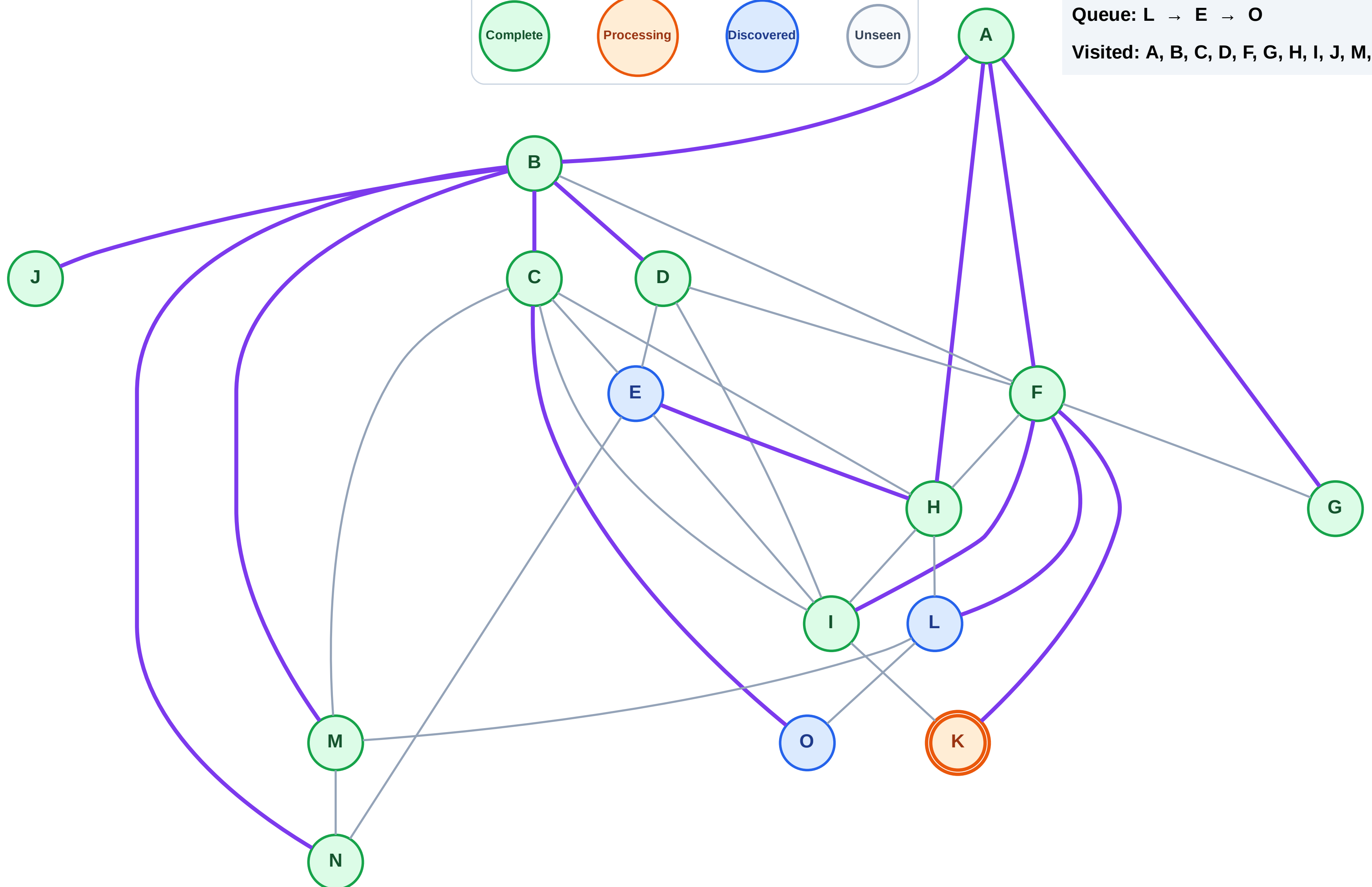
Step 24: Process node K

Legend



Queue: L → E → O

Visited: A, B, C, D, F, G, H, I, J, M, N



BFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → F → K

Legend



Complete



Processing



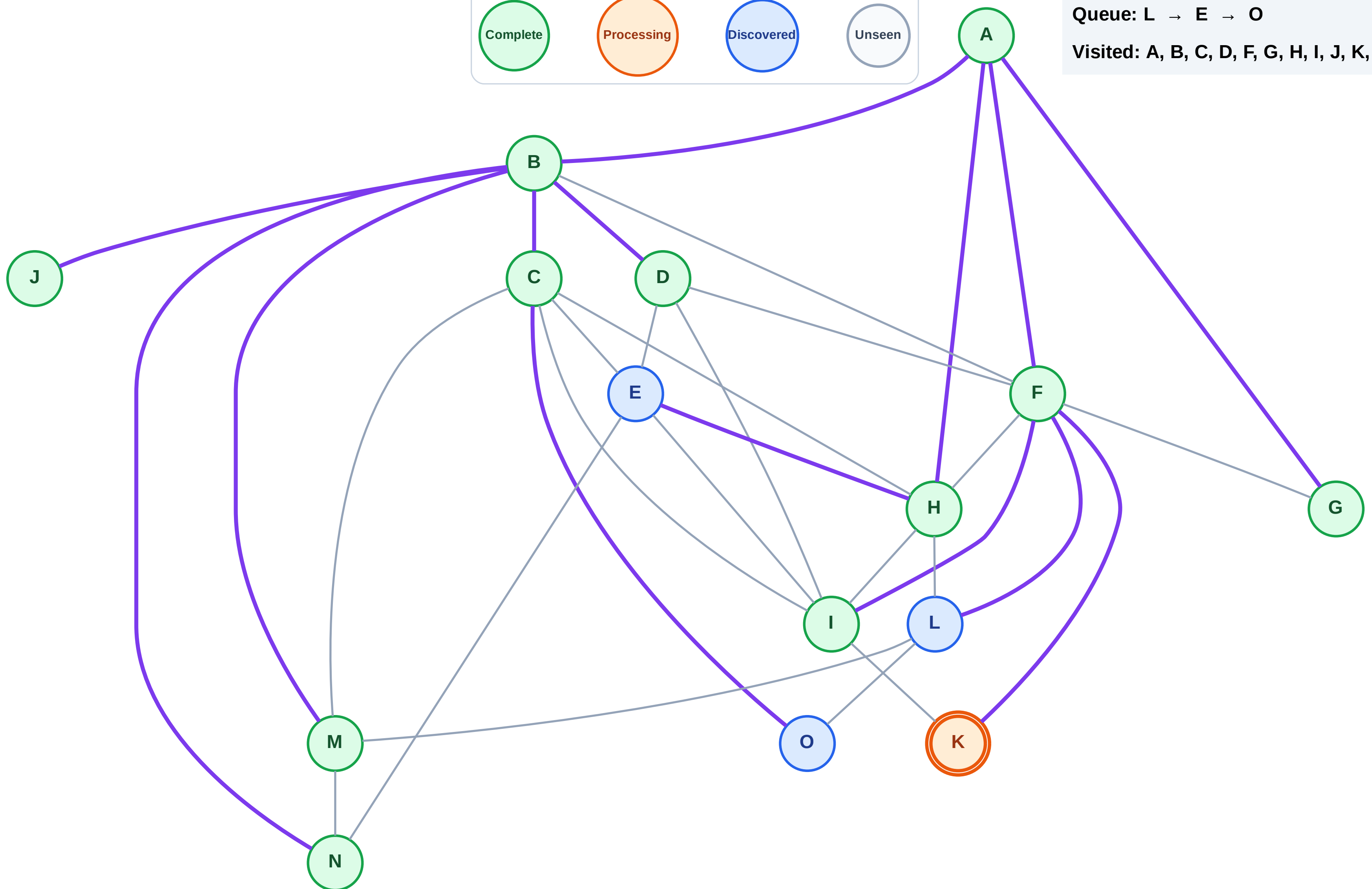
Discovered



Unseen

Queue: L → E → O

Visited: A, B, C, D, F, G, H, I, J, K, M, N



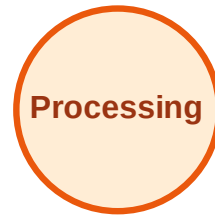
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



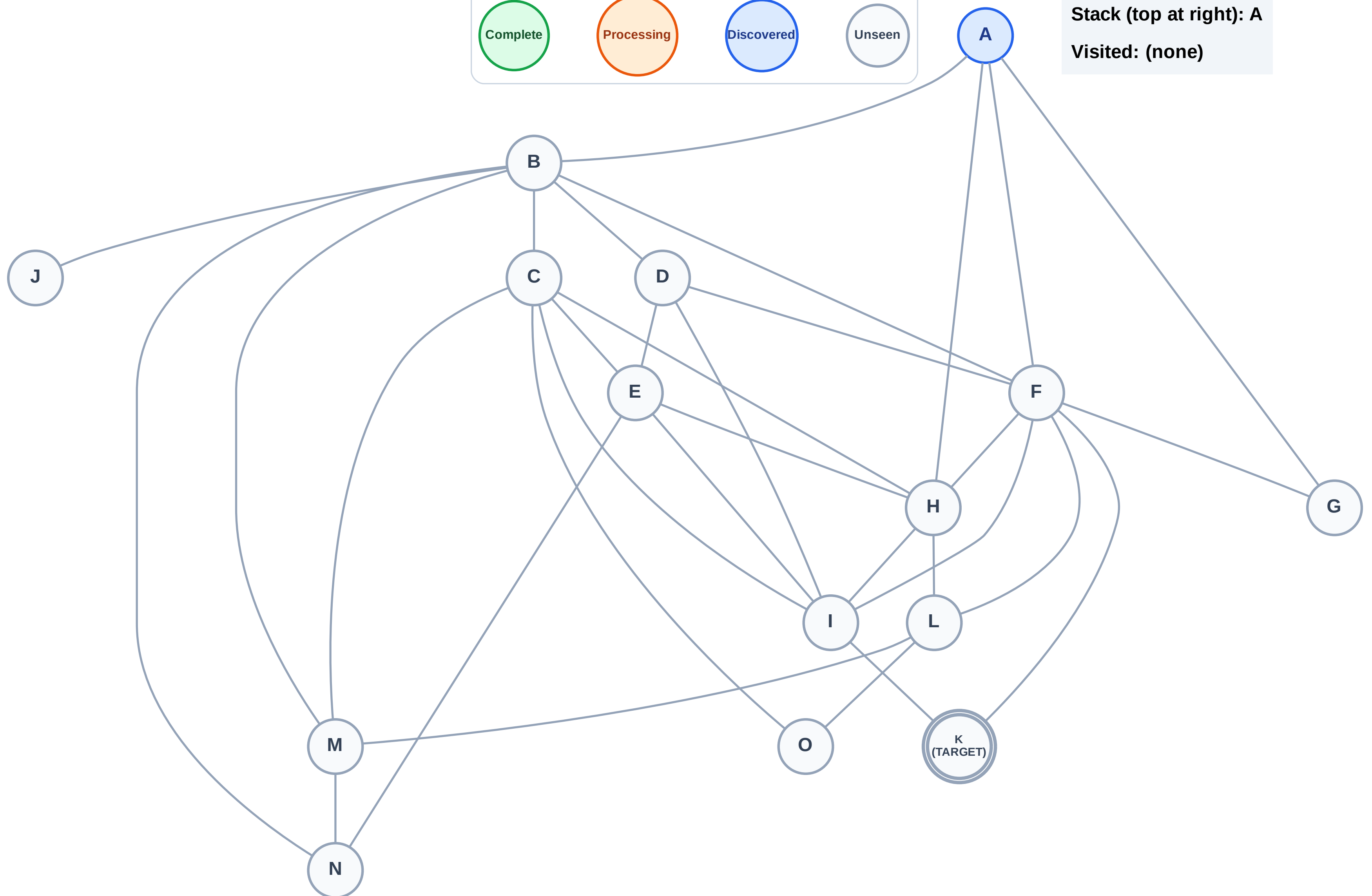
Discovered



Unseen

Stack (top at right): A

Visited: (none)



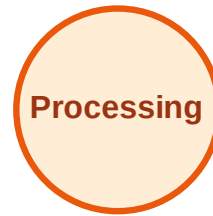
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



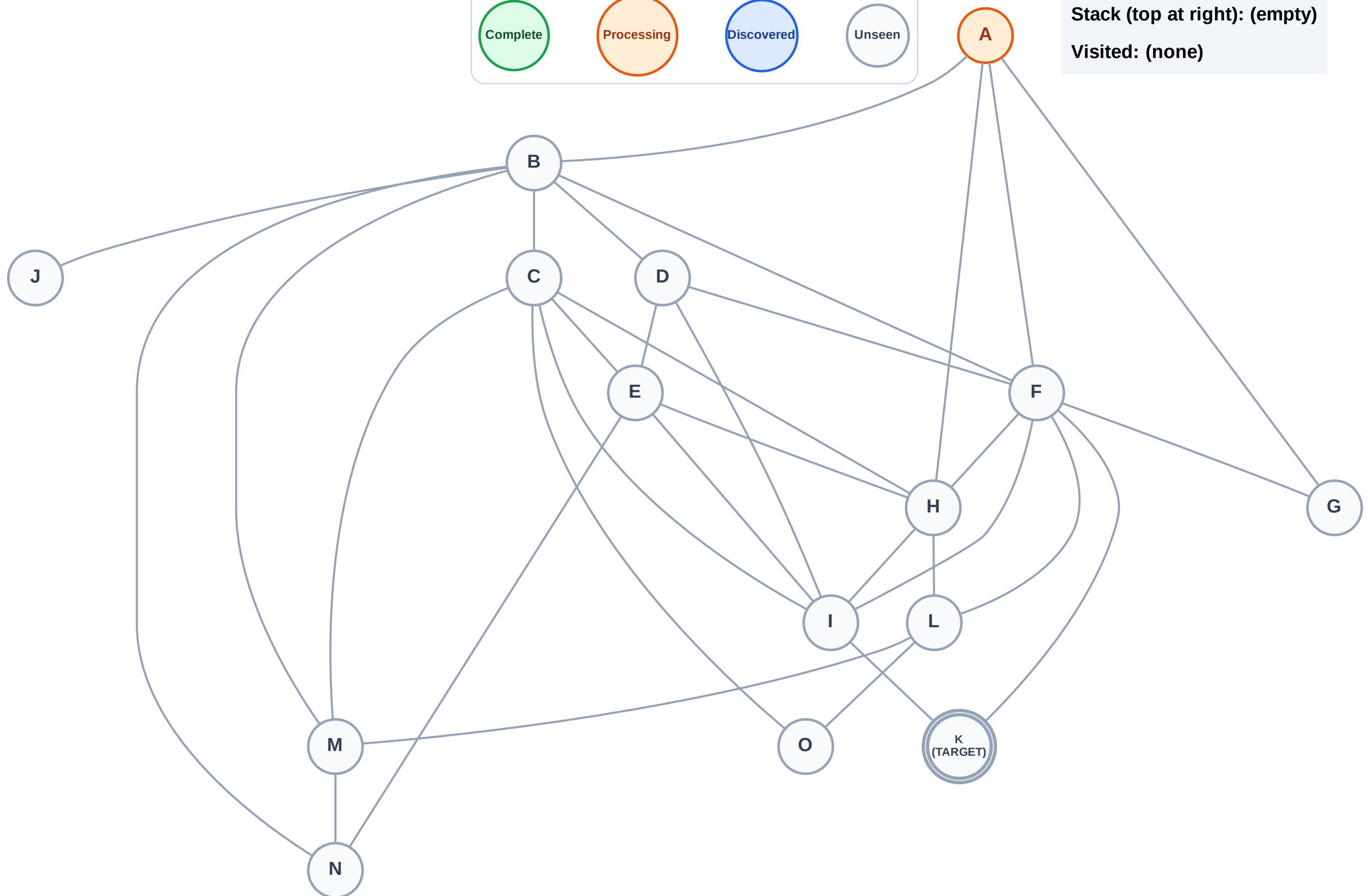
Discovered



Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

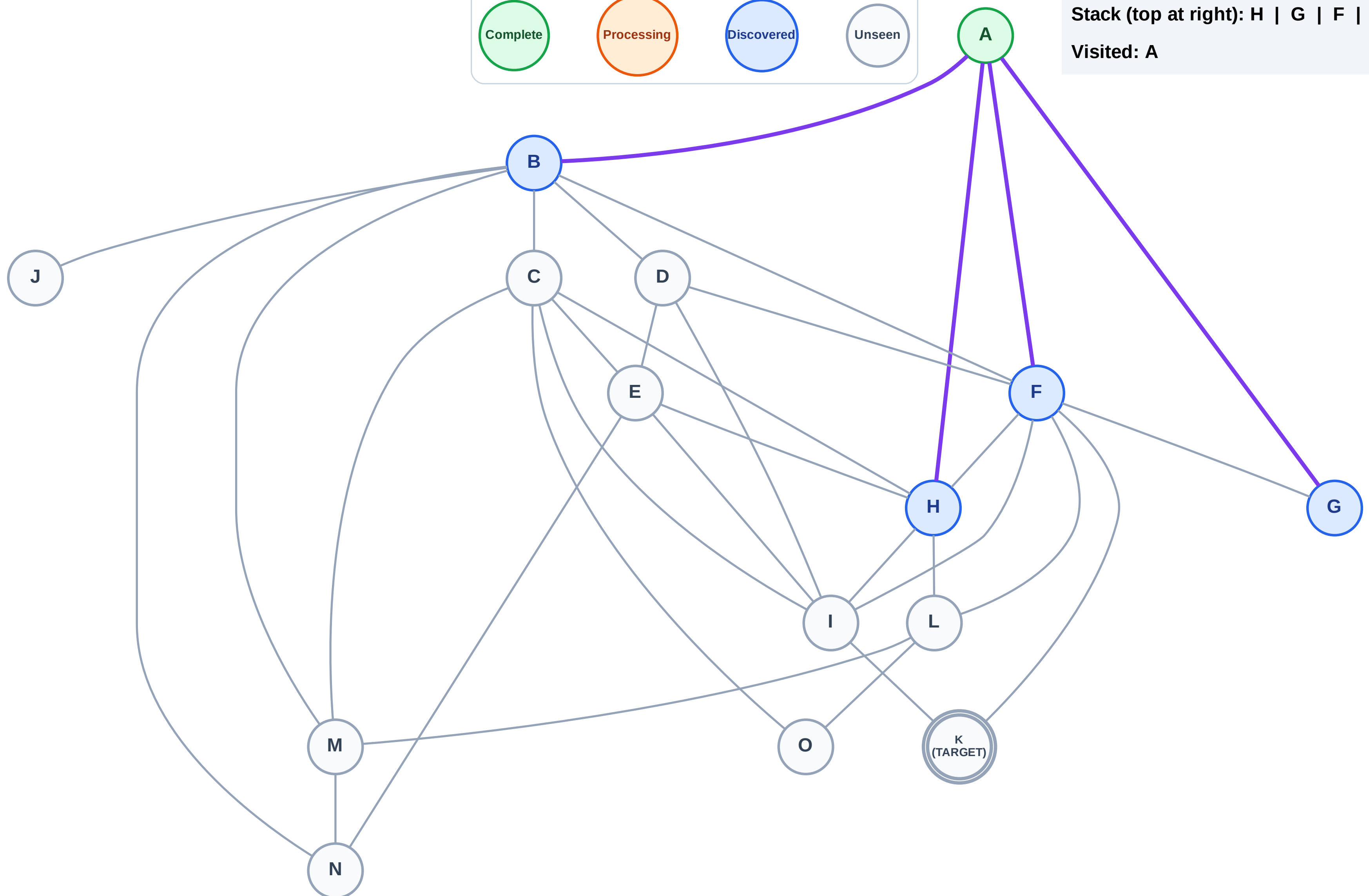
Step 3: Discover H, G, F, B from A

Legend



Stack (top at right): H | G | F | B

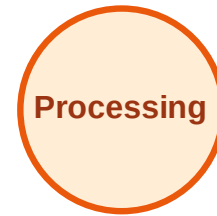
Visited: A



DFS Traversal

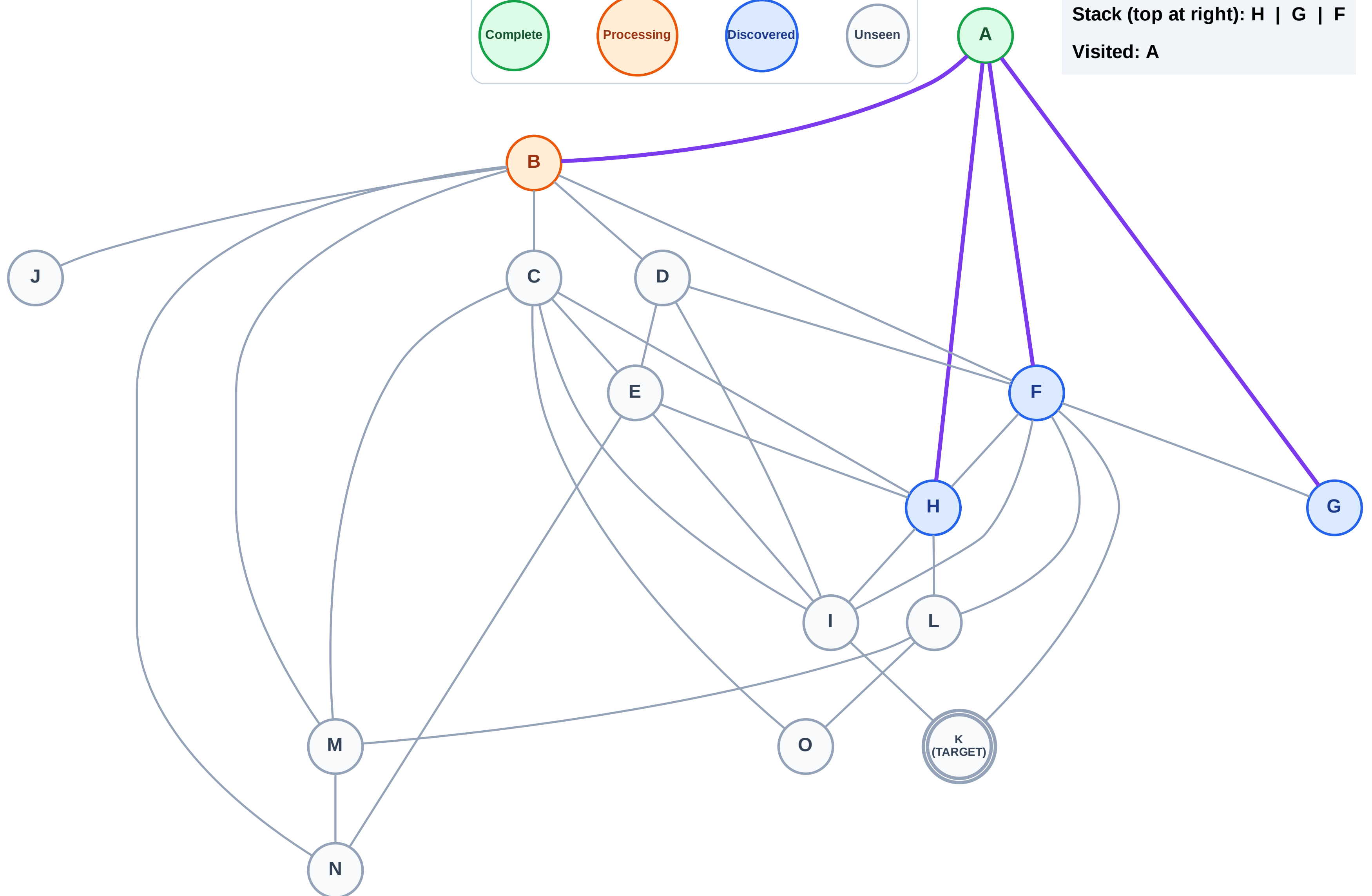
Step 4: Process node B

Legend



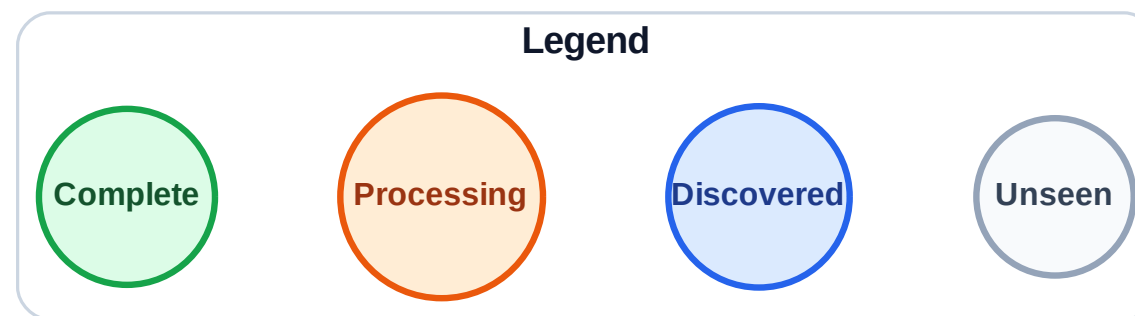
Stack (top at right): H | G | F

Visited: A



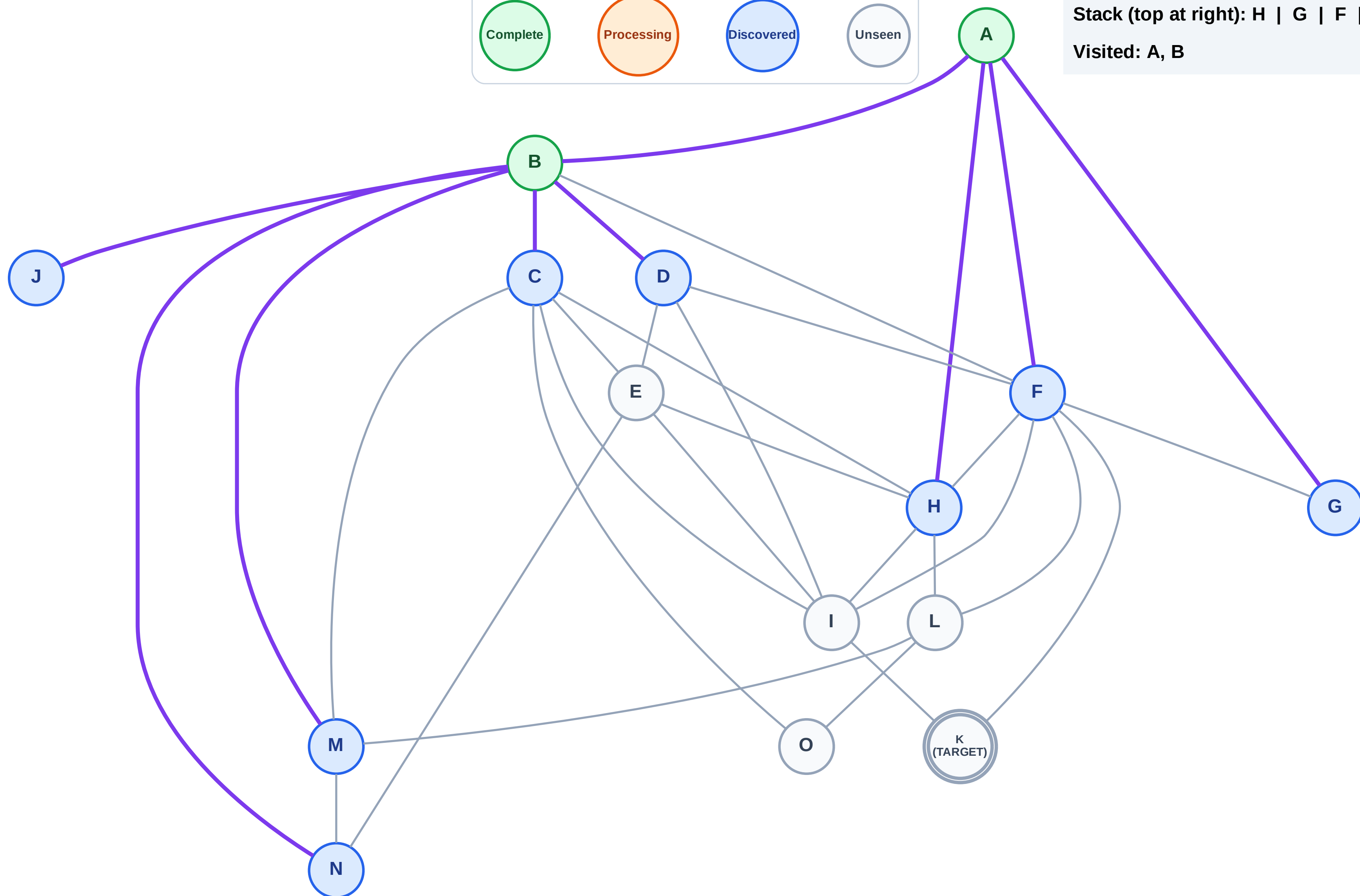
DFS Traversal

Step 5: Discover N, M, J, D, C from B



Stack (top at right): H | G | F | N | M | J | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

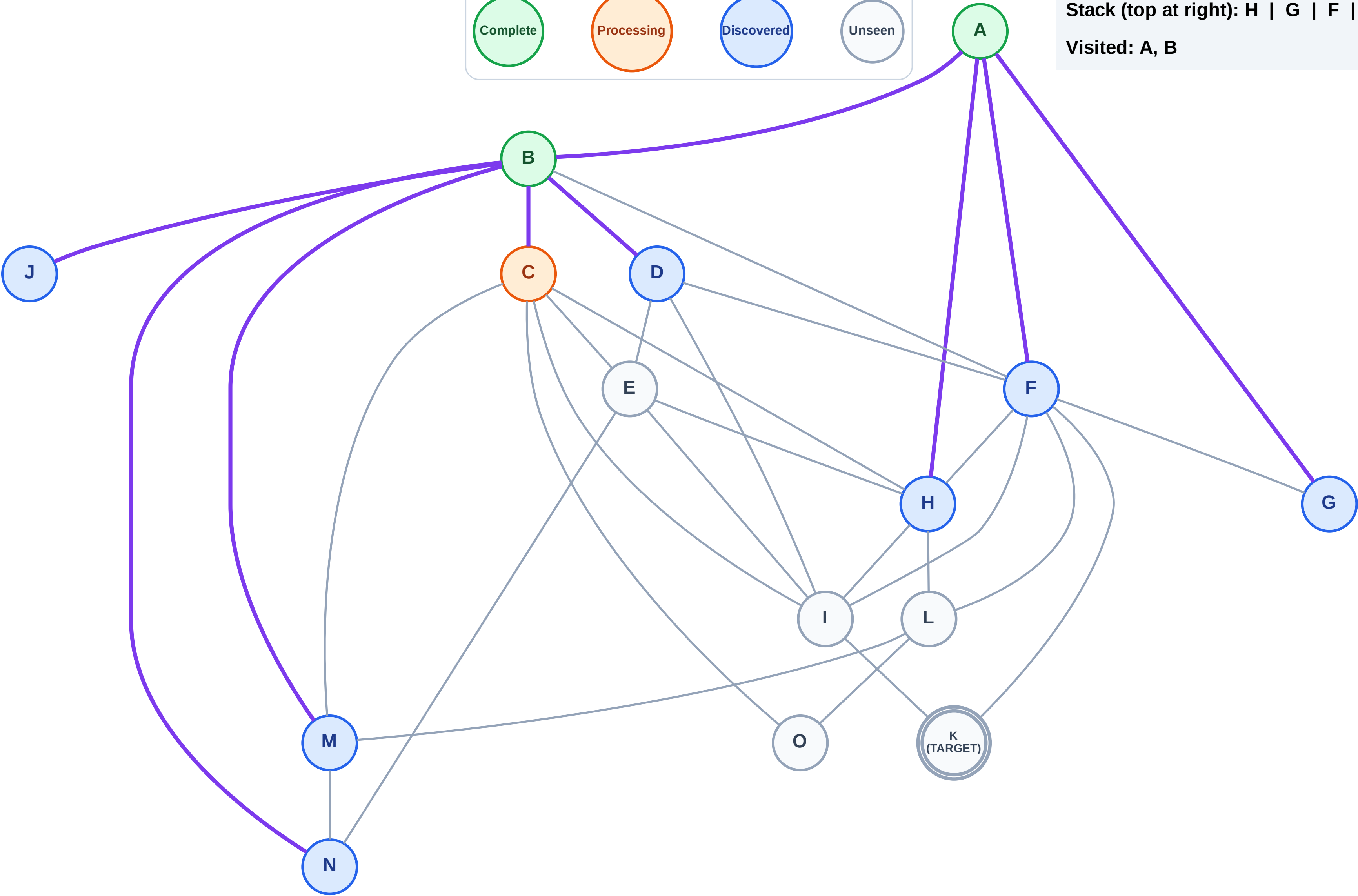
Processing

Discovered

Unseen

Stack (top at right): H | G | F | N | M | J | D

Visited: A, B



DFS Traversal

Step 7: Discover O, I, E from C

Legend

Complete

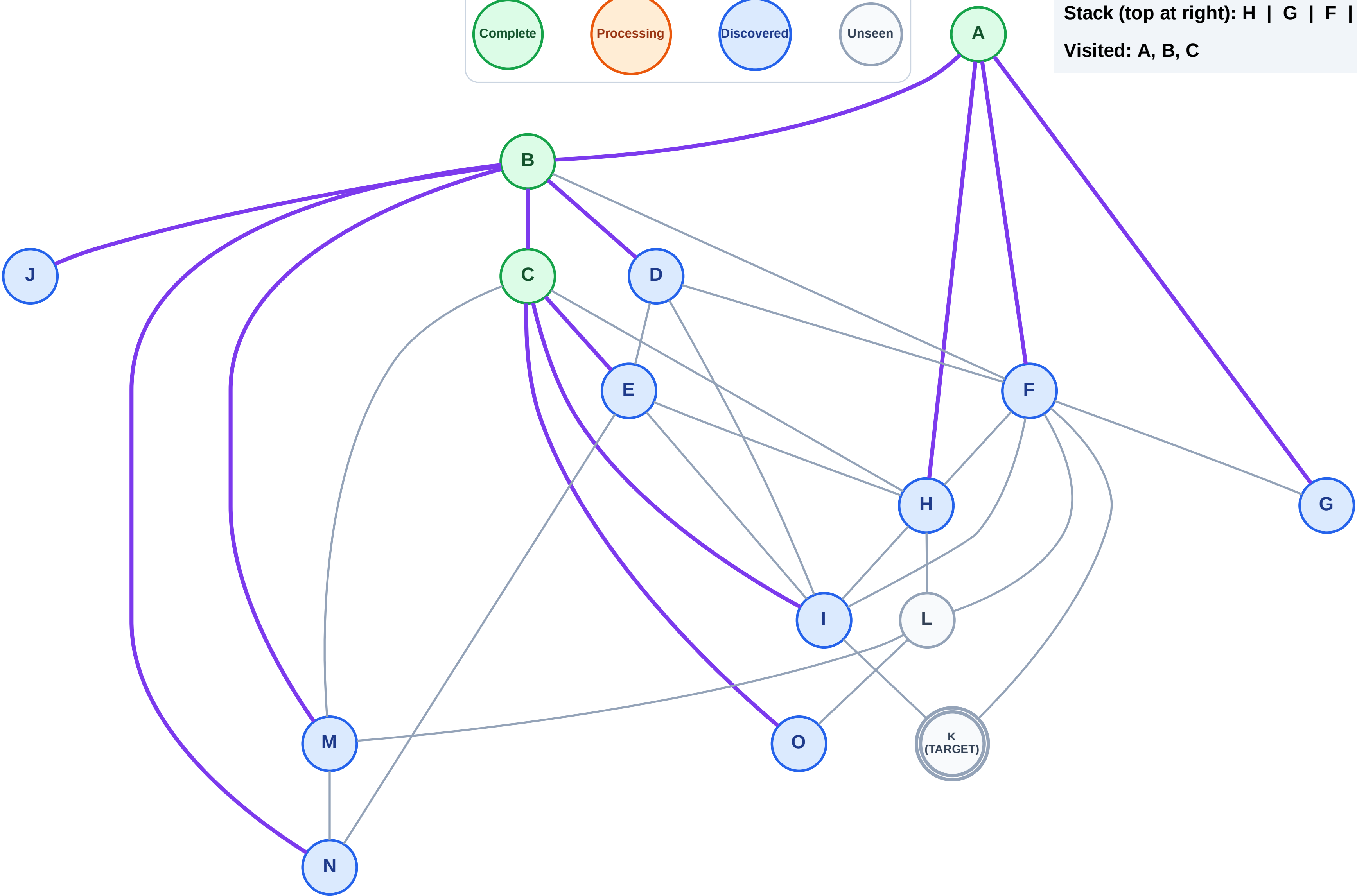
Processing

Discovered

Unseen

Stack (top at right): H | G | F | N | M | J | D | O | I | E

Visited: A, B, C



DFS Traversal

Step 8: Process node E

Legend

Complete

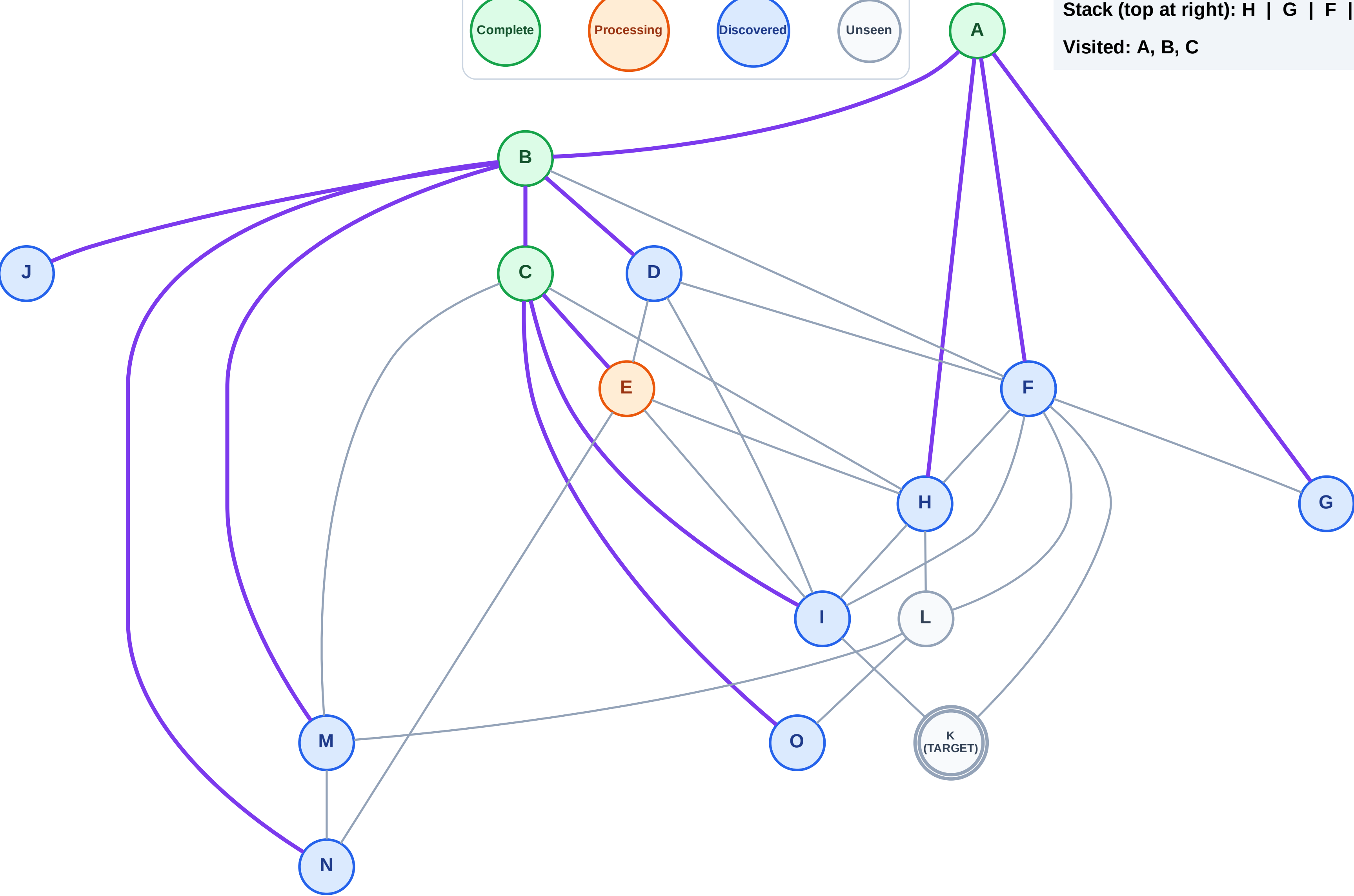
Processing

Discovered

Unseen

Stack (top at right): H | G | F | N | M | J | D | O | I

Visited: A, B, C



DFS Traversal

Step 9: No new neighbors from E

Legend

Complete

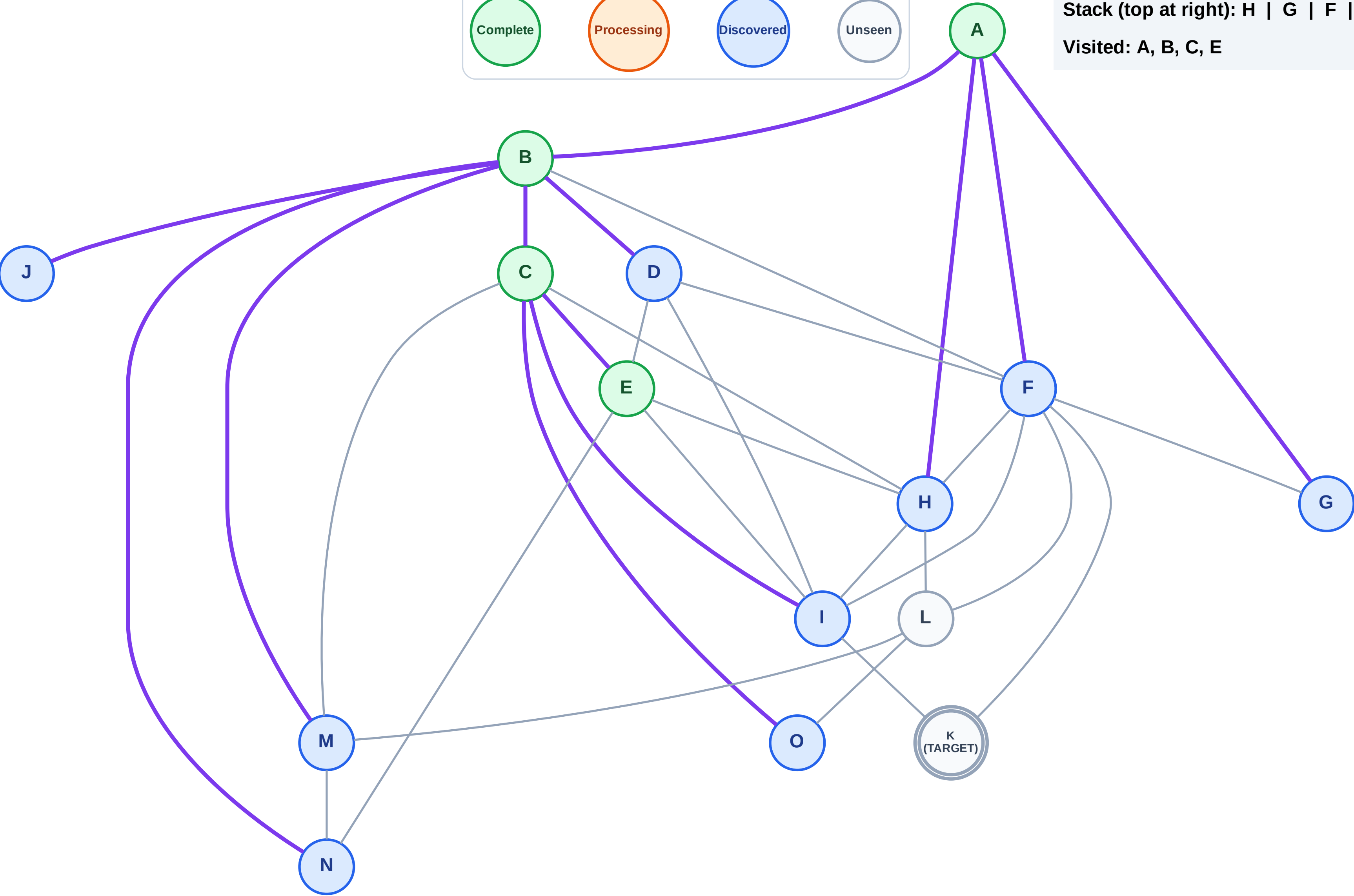
Processing

Discovered

Unseen

Stack (top at right): H | G | F | N | M | J | D | O | I

Visited: A, B, C, E



DFS Traversal

Step 10: Process node I

Legend

Complete

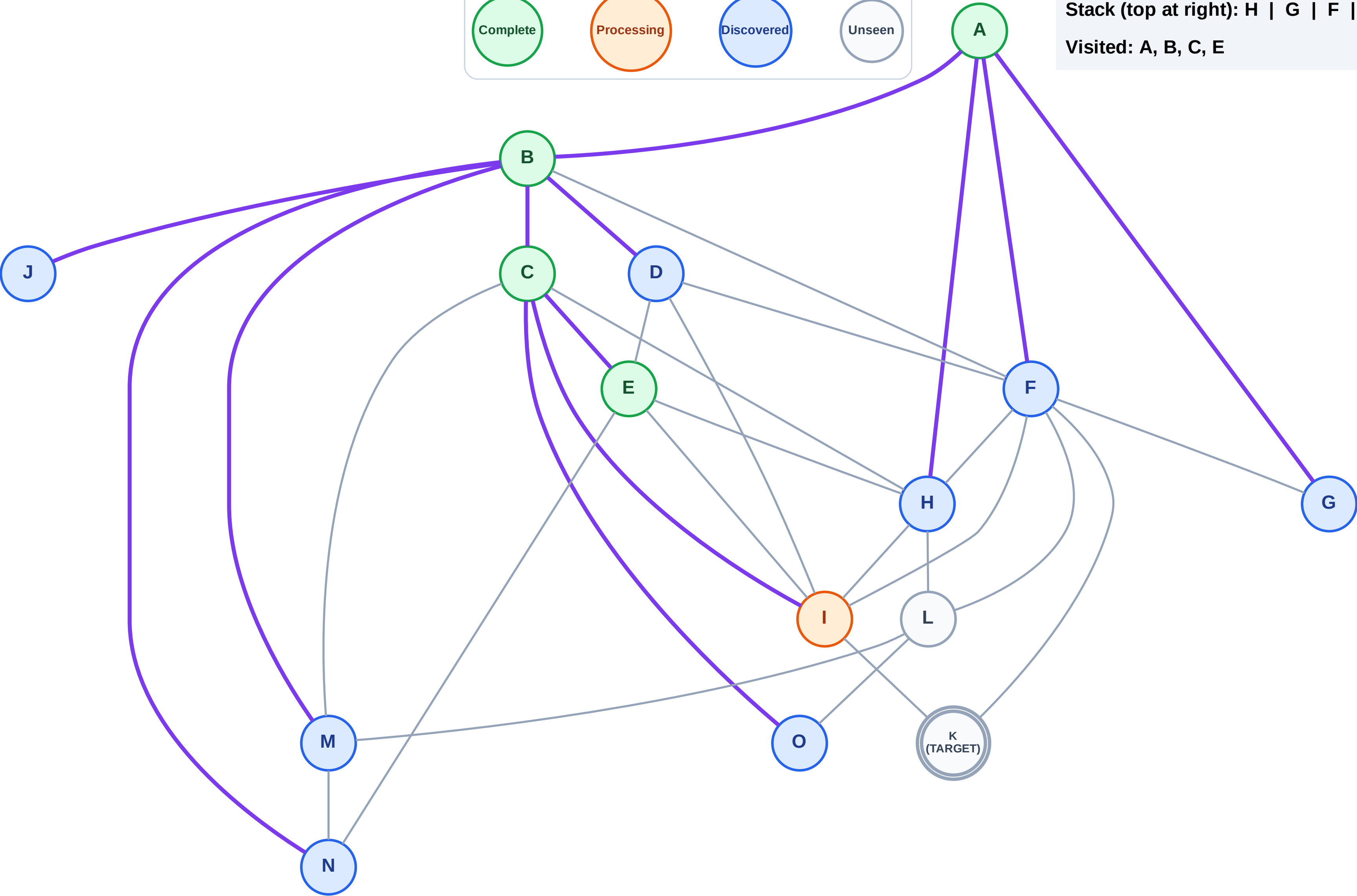
Processing

Discovered

Unseen

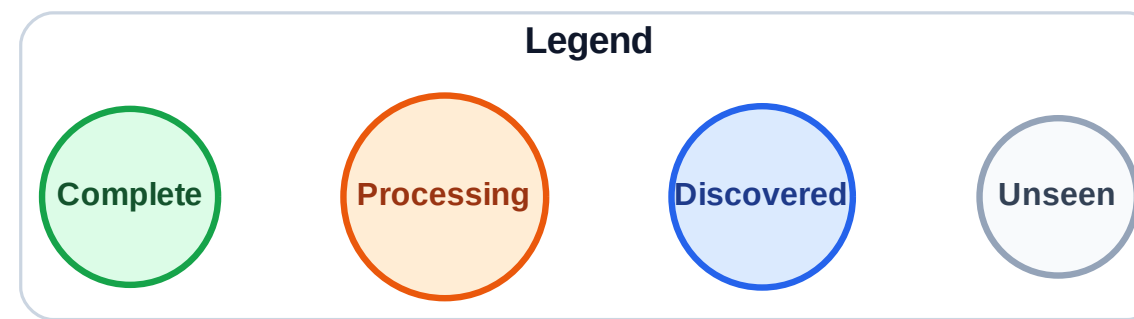
Stack (top at right): H | G | F | N | M | J | D | O

Visited: A, B, C, E



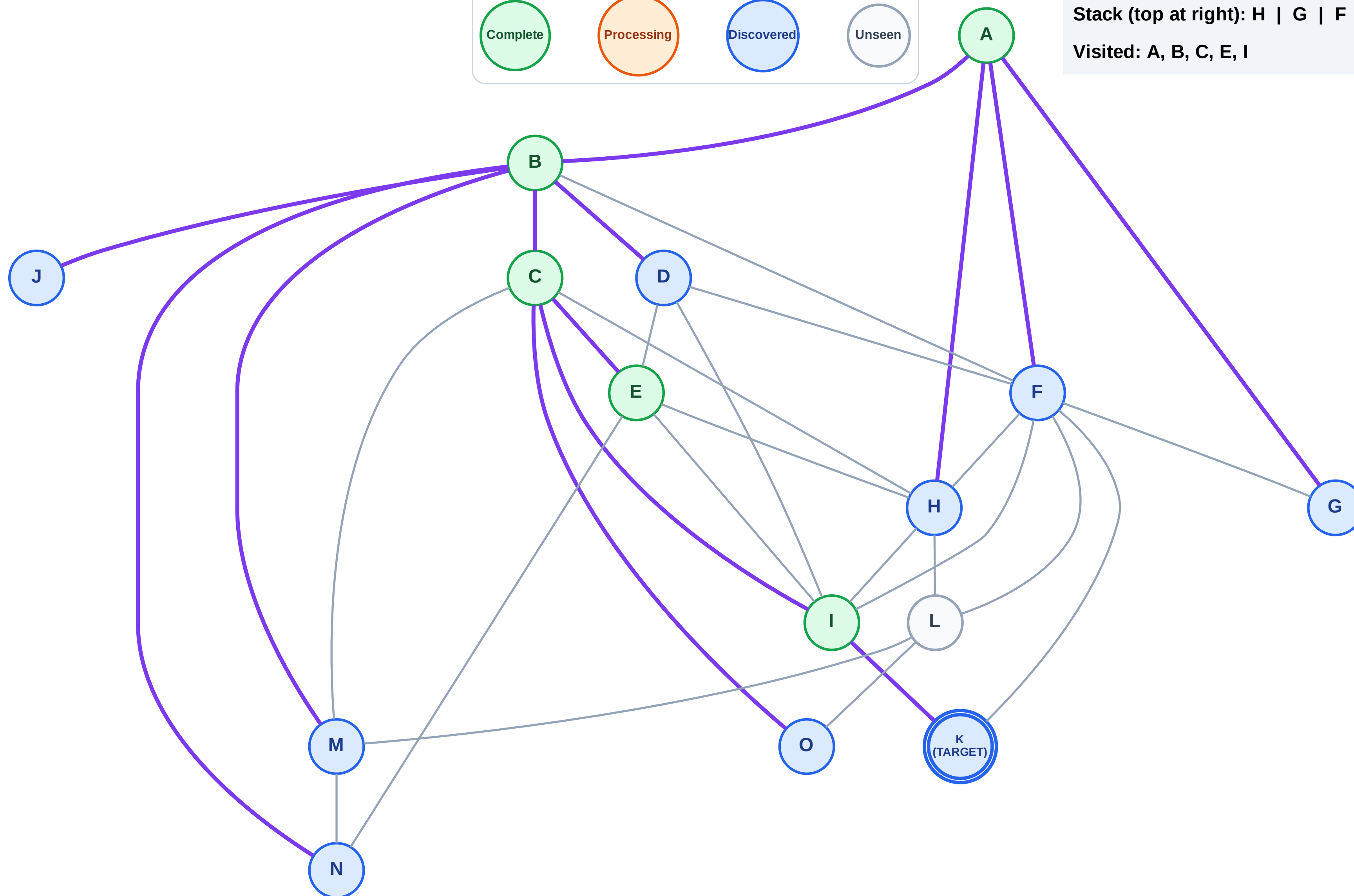
DFS Traversal

Step 11: Discover K from I



Stack (top at right): H | G | F | N | M | J | D | O | K

Visited: A, B, C, E, I



DFS Traversal

Step 12: Process node K

Legend

Complete

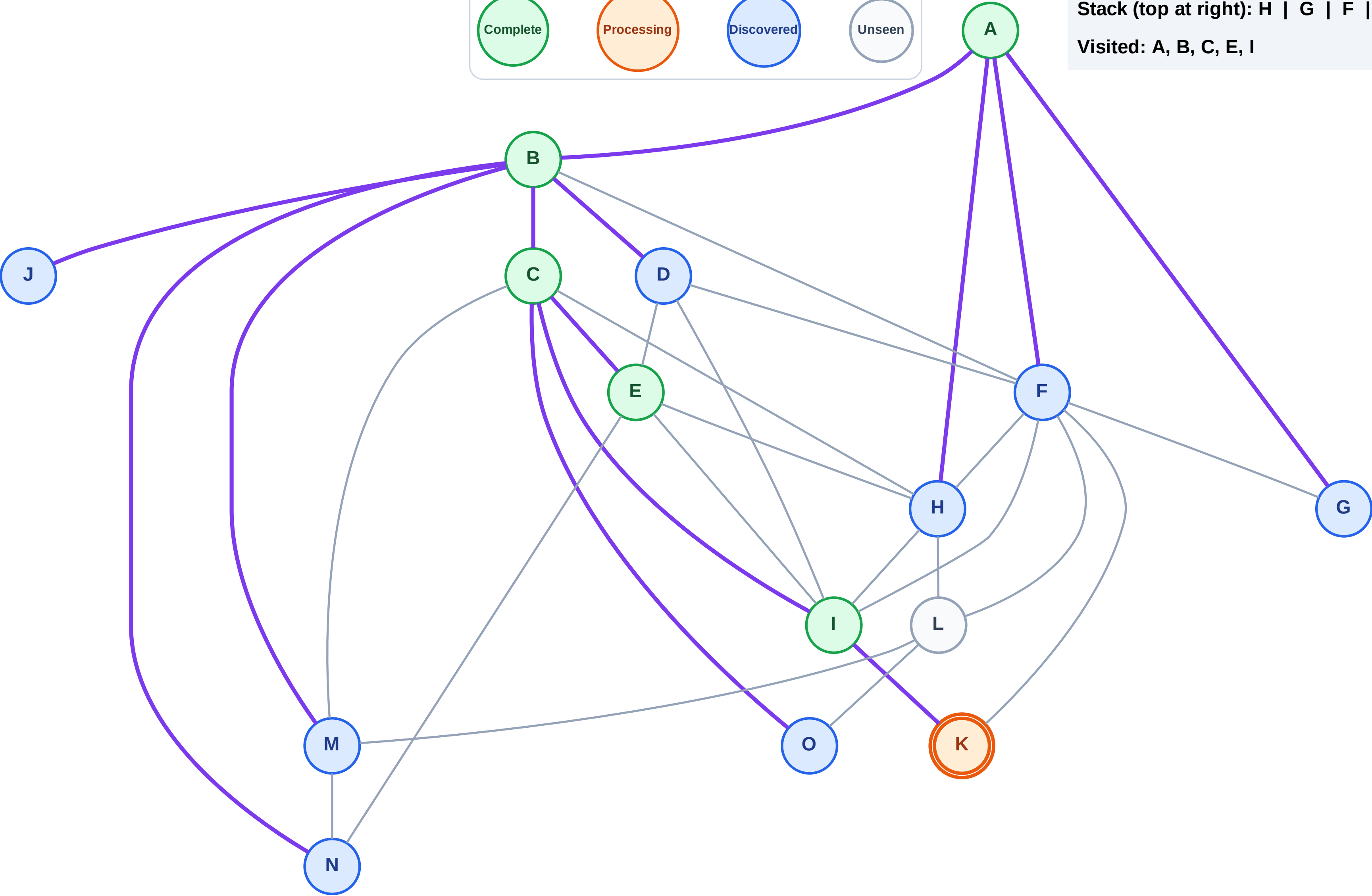
Processing

Discovered

Unseen

Stack (top at right): H | G | F | N | M | J | D | O

Visited: A, B, C, E, I



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (4 hops): A → B → C → I → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): H | G | F | N | M | J | D | O

Visited: A, B, C, E, I, K

